

How to AI an AI for Almost Anything

Lecture 1 - Introduction

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MIT Media Lab & MIT EECS



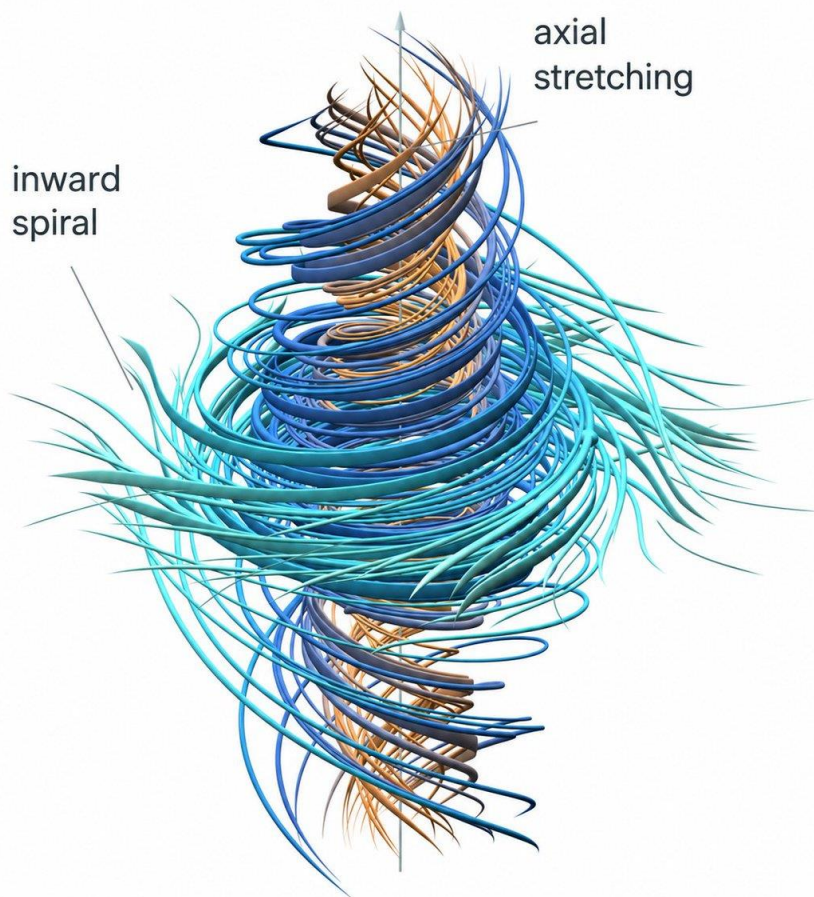
<https://pliang279.github.io>

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 [@pliang279](#)



AI for Discovery



“Our internal model group arrived at the Navier–Stokes solution in 88 hours, using around 10,000 coordinating AI agents.”



Noam Brown ✓
@polynoamial



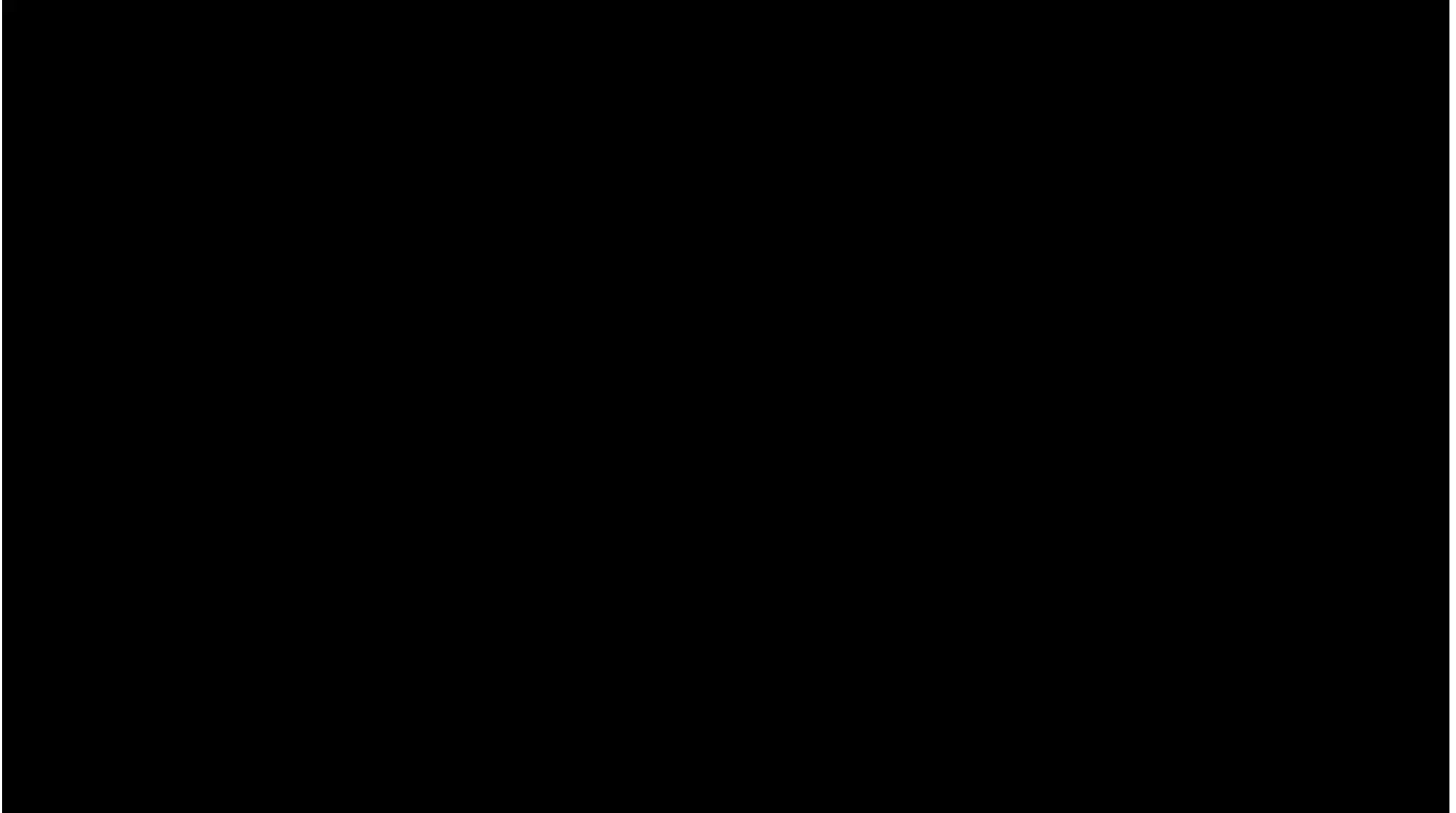
Yes, this result cost millions of dollars.

But remember that when @OpenAI announced o3 it cost ~\$500,000 to score 87.5% on ARC-AGI 1. Today, Astra scores higher for ~\$20.

In 2025 it took us and GDM an enormous amount of compute to achieve IMO gold. For the 2026 IMO, anyone with a \$20/month ChatGPT subscription could do it.

Massively scaling test-time compute gives us a glimpse of the future. I believe that a year from now everyone will have an AI at their fingertips capable of solving problems of this caliber.

Computer Use Agents



New Themes in AI

Test-time scaling

AI agents + multi-agent

User interaction and rich feedback

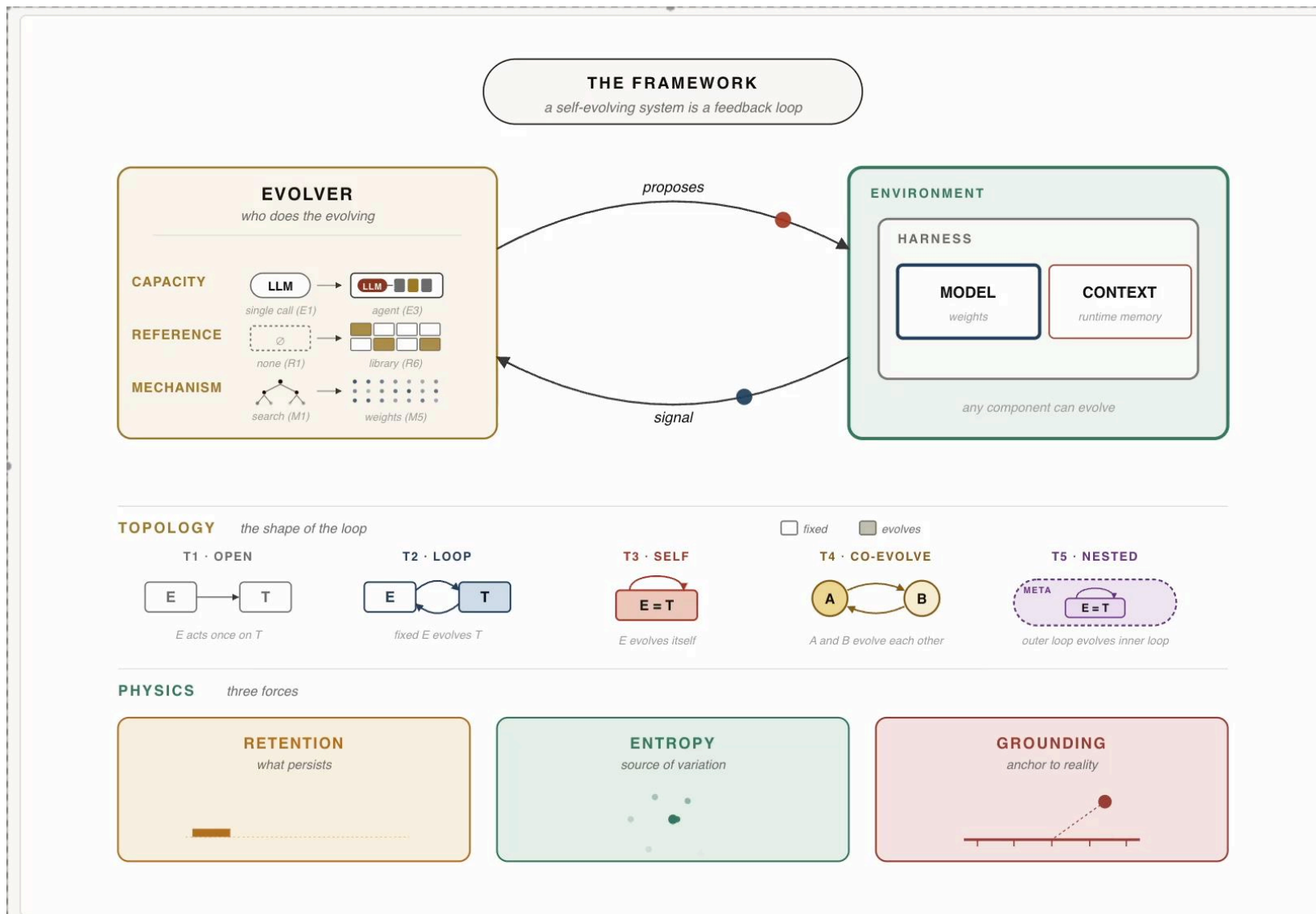
Continual learning

Tool use and long-term memory

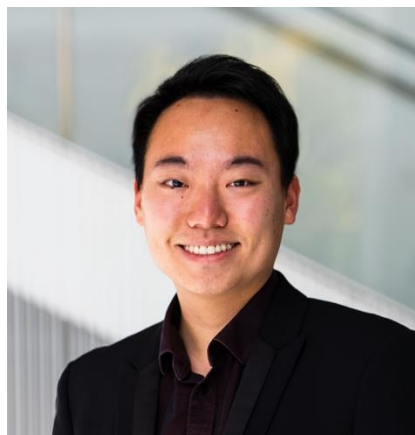
Reinforcement learning and environment design

Self-Evolving AI

<https://quao627.github.io/blog/self-evolving-agents/>



Your Teaching Team, Spring 2026



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Course instructor



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Teaching Assistant

Learning Objectives

- 1 Study recent technical achievements in AI research
- 2 Improve critical and creative thinking skills
- 3 Understand future research challenges in AI
- 4 Explore and implement new research ideas in AI

Preferred Pre-requisites

- 1 Knowledge of AI programming and experimentation
- 2 Understanding of modern AI capabilities & limitations
- 3 Bring external (non-AI) domain knowledge about your problem
- 4 Bonus: worked on self-evolving AI, AI for discovery, etc.

Course delivery format

- 2-hour class every Wednesday, 2 - 4pm.
- Homework and reading assignments outside of class, 2-week duration.
- Significant research project outside of class, with reports and presentations.

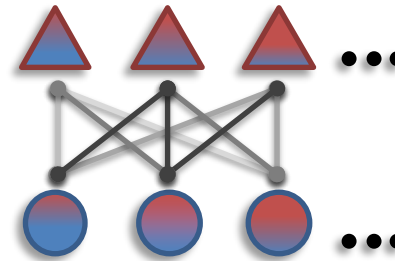
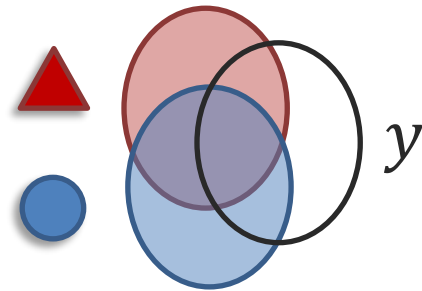
Lecture Topics *(subject to change, based on student interests and course discussions)*

Module 1: Context evolution

Week 2 (9/16): Prompt optimization, memory, tools, knowledge

Week 3 (9/23): LLMs as optimizers for discovery

Week 4 (9/30): Project proposal presentations



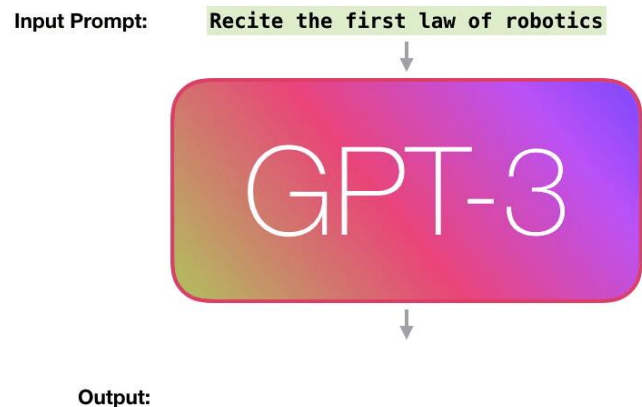
Lecture Topics (subject to change, based on student interests and course discussions)

Module 2: Harness evolution

Week 5 (10/7): Harness components, multi-agent systems

Week 6 (10/14): Self-evolving harnesses

Week 7 (10/21): Project midterm presentations



*An armchair in
the shape of an
avocado*

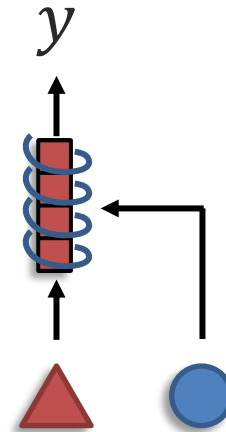
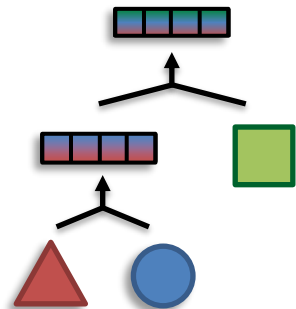


Lecture Topics *(subject to change, based on student interests and course discussions)*

Module 3: Model evolution

Week 8 (10/28): Continual learning, distillation, test-time training

Week 9 (11/4): Self-play approaches



Lecture Topics *(subject to change, based on student interests and course discussions)*

Module 4: Environment evolution

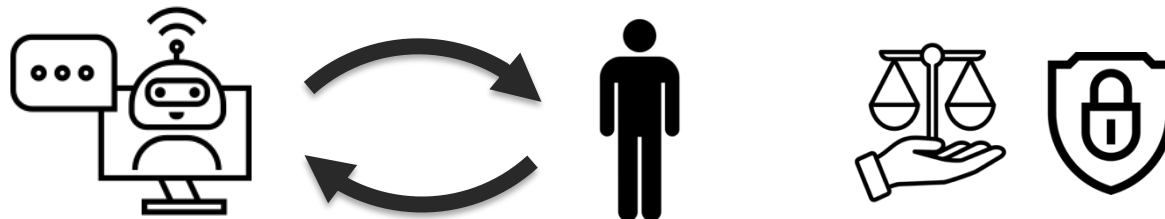
Week 10 (11/11): Synthetic data and tasks, environment generation

Week 11 (11/18): World models, evolving evaluation and rewards

Week 12 (11/25): Future topics – AI for discovery, multimodal agents

Week 13 (12/2): Project final presentations

Week 14 (12/9): Future topics – Embodied harnesses, safe evolution (NeurIPS)



Grading Overview

- 40% for homework assignments:
 - 4 homework assignments, roughly 2 weeks for each.
- 20% for reading assignments:
 - 4 reading assignments covering state-of-the-art research in multimodal AI, released with the homework, roughly 2 weeks for each, with discussion/presentations in class.
- 40% for a high-quality research project:
 - 10% for proposal with literature review
 - 10% for midterm progress report
 - 20% for final report and presentation

Homework Assignments

Homeworks will involve implementing and applying some key algorithms:

HW 1, due week 4: Context evolution.

- Tentatively on prompt optimization (TextGrad) and using LLMs to optimize (OpenEvolve).

HW 2, due week 7: Harness evolution.

- Tentatively on skill evolution (SkillOpt) and self-generating harness (Meta-Harness).

HW 3, due week 10: Model evolution.

- Tentatively on test-time training and continual learning methods.

HW 4, due week 12: Task evolution.

- Tentatively on self-generating tasks and putting everything together on PostTrainBench.

Reading Assignments

- 4 readings assignments, with usually 2 required papers and some suggested (but optional) papers, and 5-6 discussion probes.
- Three main assignment parts:
 - **Reading notes:** Read the assigned papers and summarize the main take-away points
 - *Optional: if you have clarification questions about the papers*
 - **Paper scouting:** Scout for extra papers, blog posts or other resources related to these question probes
 - **Discussion points:** Reflect on the question probes related to the reading papers, prepare discussion points, with live discussion and brainstorming in class.
- Goal is to jointly write a survey paper on self-evolving AI combining each discussion session, focusing on a clear taxonomy and rigorous new directions.

Course Project

- Meant to introduce students to the research process:
 - Reading literature
 - Becoming familiar with application domains
 - Trying state-of-the-art models
 - Performing error analysis
 - Developing and iterating on new methods
 - Evaluating and analyzing results
- Done in groups of 2-4 students.
- Group will be assigned a mentor.
- Equal contribution tracked via regular updates, presentations, and GitHub.

Course Project Timeline

- Week 2: **Project preferences:** you should have selected your teammates, have ideas about your dataset and task
- Week 4: **Proposal report + presentation:** literature review & research ideas
- Week 5: Setting up datasets and baselines
- Week 7: **Midterm report + presentation:** baselines & results for new ideas
- Week 10: Updated results for research idea, analysis, ablations
- Week 13: **Final presentations:** describing explored research ideas with results, analysis, discussion
- Week 14: **Final report**

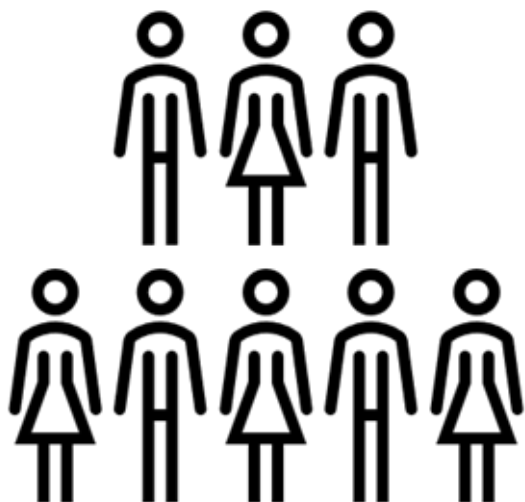
Absences and Late Submissions

- Students expected to attend classes
 - If you plan to miss more than one lecture this semester, let us know as soon as possible.
 - Course participation and in-class discussions are a graded component (20%).
- Homework and reading assignment wildcards (2 per student)
 - 24-hours extension, max 1 per assignment
- Project report wildcards (2 per team)
 - Either for proposal, midterm, or final report
 - 24-hours extension, can be used together

Course Websites

- Course website: <https://mit-mi.github.io/how2ai-course/fall2026/>
- Canvas for submissions: <https://canvas.mit.edu/courses/40521/>

Time for Introductions!



Your name, department and programs

Why self-evolving AI?

Previous research experience and applications

Why are you interested in this course?

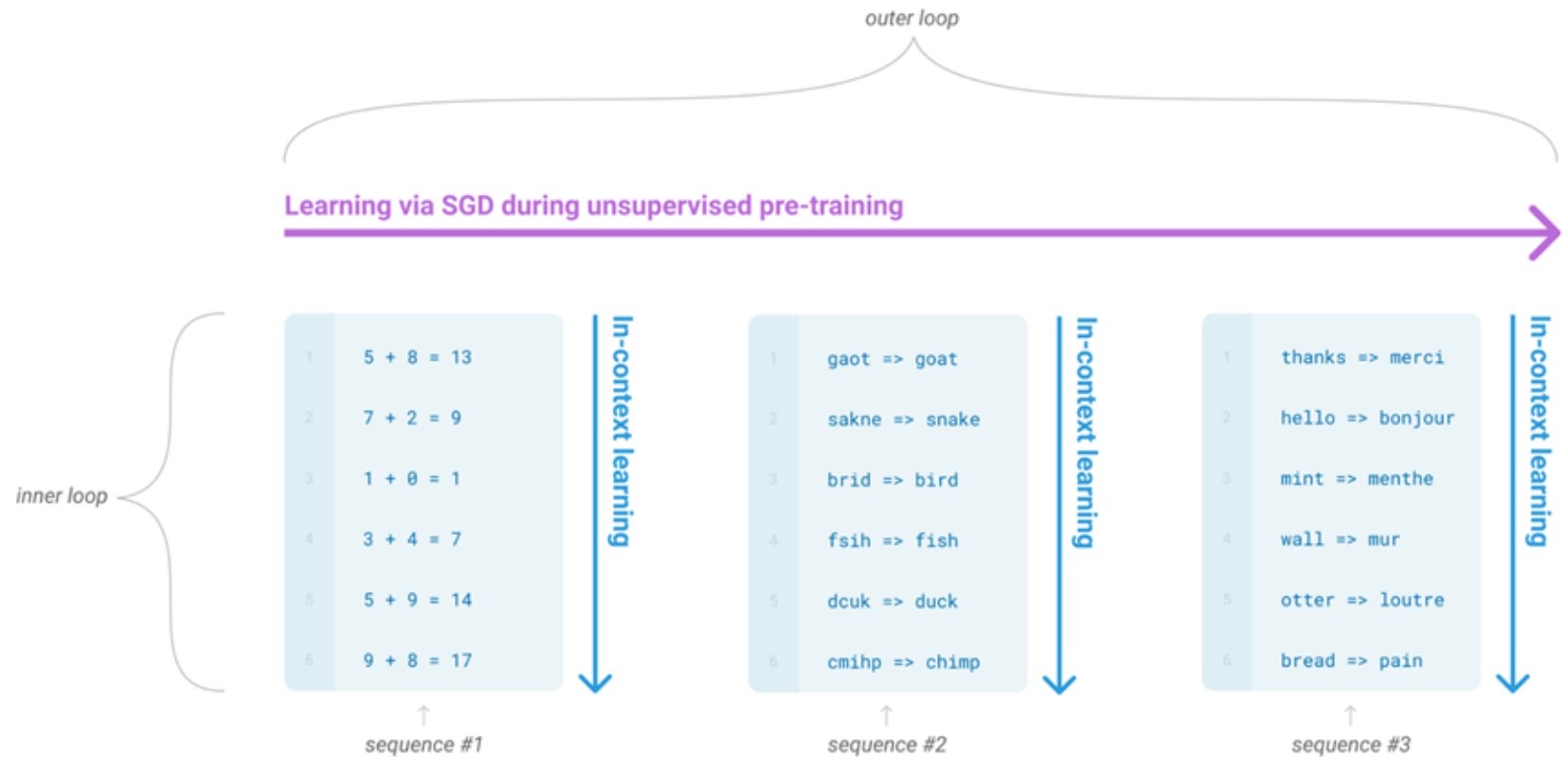
Today's Lecture

- Background to LLMs, reasoning, and agents
- Introduction to self-evolving AI
- Research ideas and tasks for self-evolving AI

From Language Models to Autonomous Agents: Six Years of Progress

2020 In-Context Learning

GPT-3 demonstrates zero-shot and few-shot task adaptation after large-scale unsupervised pre-training



2020 Few-Shot Prompting (GPT-3)

Zero-shot

```
Translate English to  
French:  
cheese => fromage
```

One-shot

```
Translate English to  
French:  
sea otter => loutre de mer  
cheese => fromage
```

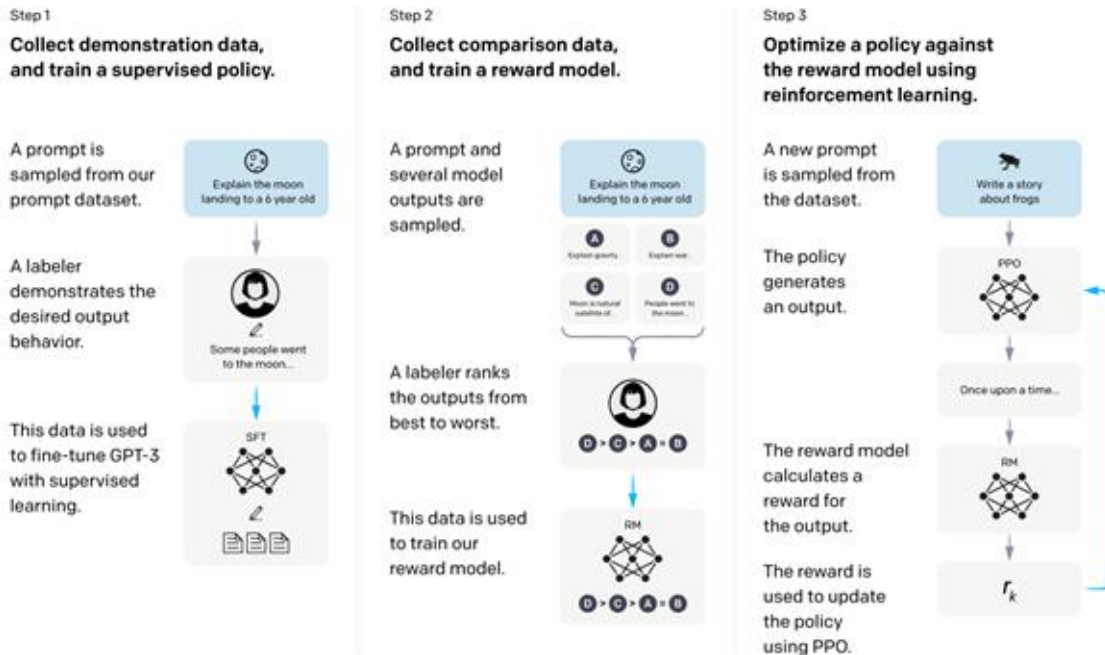
Few-shot

```
Translate English to  
French:  
sea otter => loutre de mer  
peppermint => menthe  
poivrée  
plush girafe => girafe  
peluche  
cheese => fromage
```

GPT-3 demonstrates zero-shot and few-shot task adaptation.

2022 Instruction following

Instruction tuning and RLHF enable LLMs to better follow user instructions



2022 GPT-3 vs InstructGPT on the same prompt

Explain the moon landing to a 6 year old in a few sentences.

GPT-3

Explain the theory of gravity to a 6 year old.
Explain the theory of relativity to a 6 year old in a few sentences.
Explain the big bang theory to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

GPT-3 continues the text. InstructGPT, after supervised fine-tuning and RLHF, answers the question.

2022 Chain-of-thought reasoning

Chain-of-thought prompting elicits multi-step reasoning in large language models

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

2022 Chain-of-thought reasoning

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

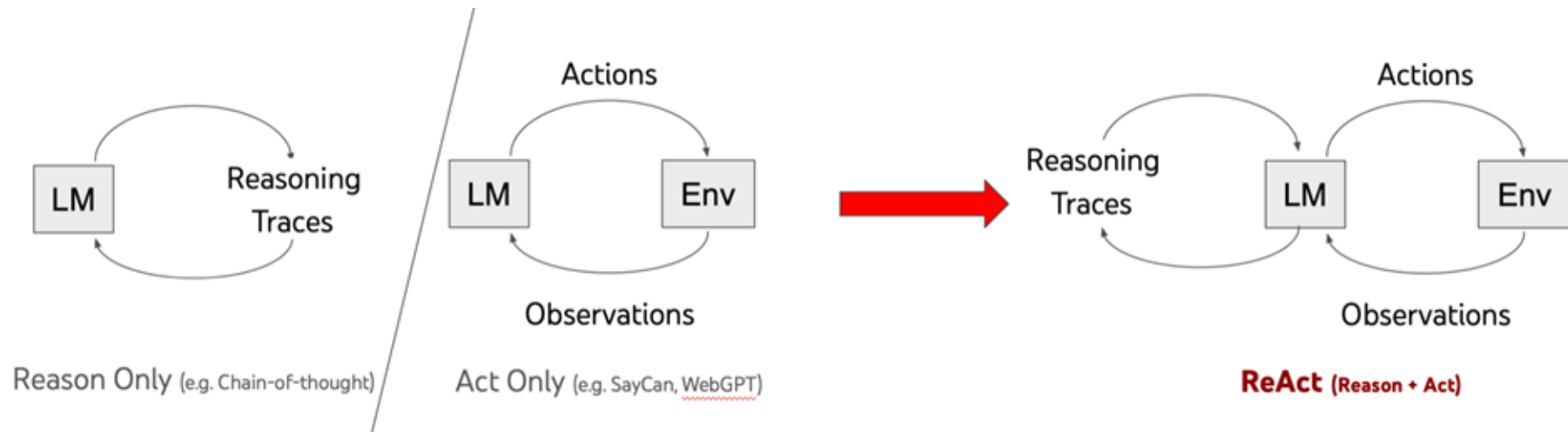
A: **Let's think step by step.**

(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓*

A simple prompt can also bemused to elicit chain-of-thought reasoning

2023 Reasoning and acting

LLMs interleave reasoning, tool use, and environment interaction



2023 Reasoning and acting

Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

Thought 1 I need to search Apple Remote and find the program it was designed for.

Act 1 Search[Apple Remote]

Obs 1 ... originally designed to control the Front Row media center program ...

Thought 2 Now I need to find what else can control Front Row.

Act 2 Search[Front Row (software)]

Obs 2 Front Row ... controlled by an Apple Remote or the keyboard function keys.

Act 3 Finish[keyboard function keys] ✓

Each thought guides the next search. Each observation motivates the next thought.

2024-2025 RL-trained reasoning

Reinforcement learning improves complex reasoning and enables models to benefit from additional test-time computation

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

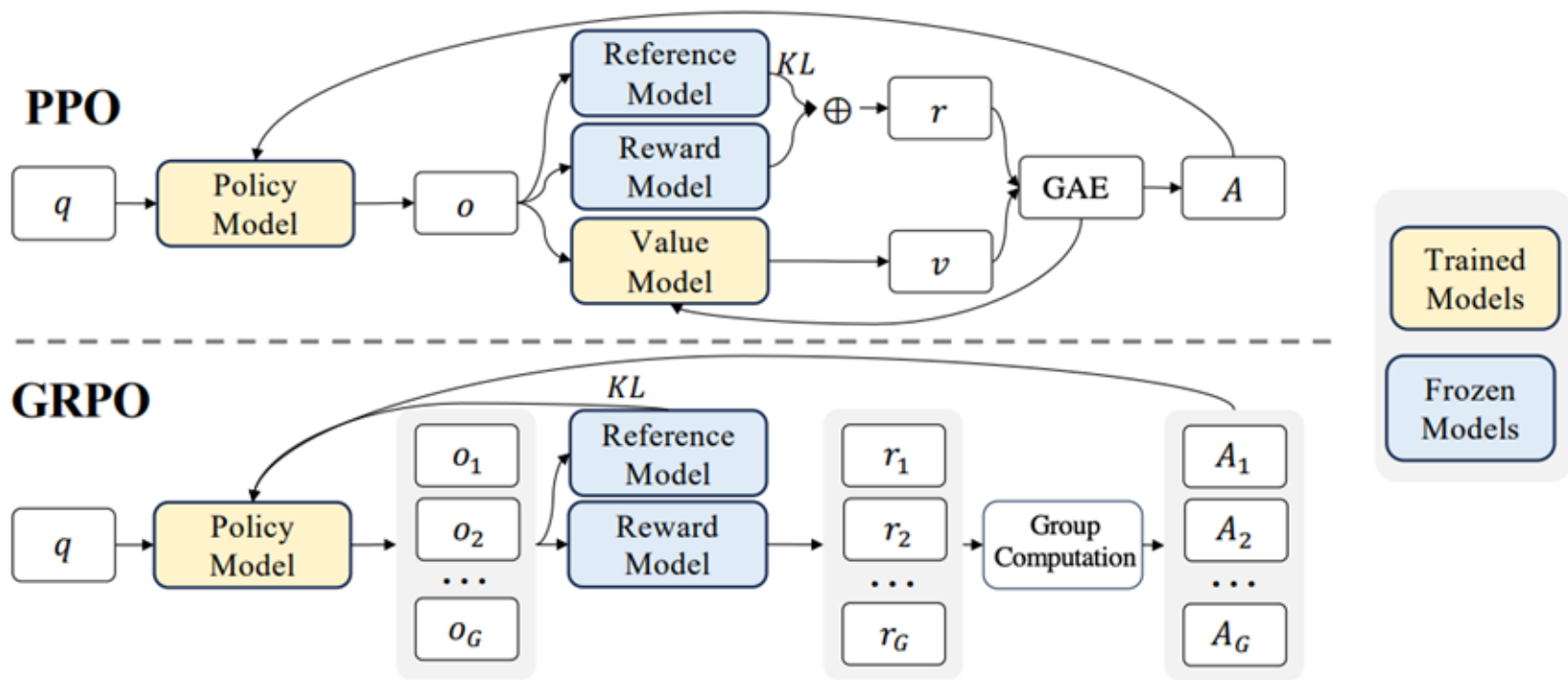
$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

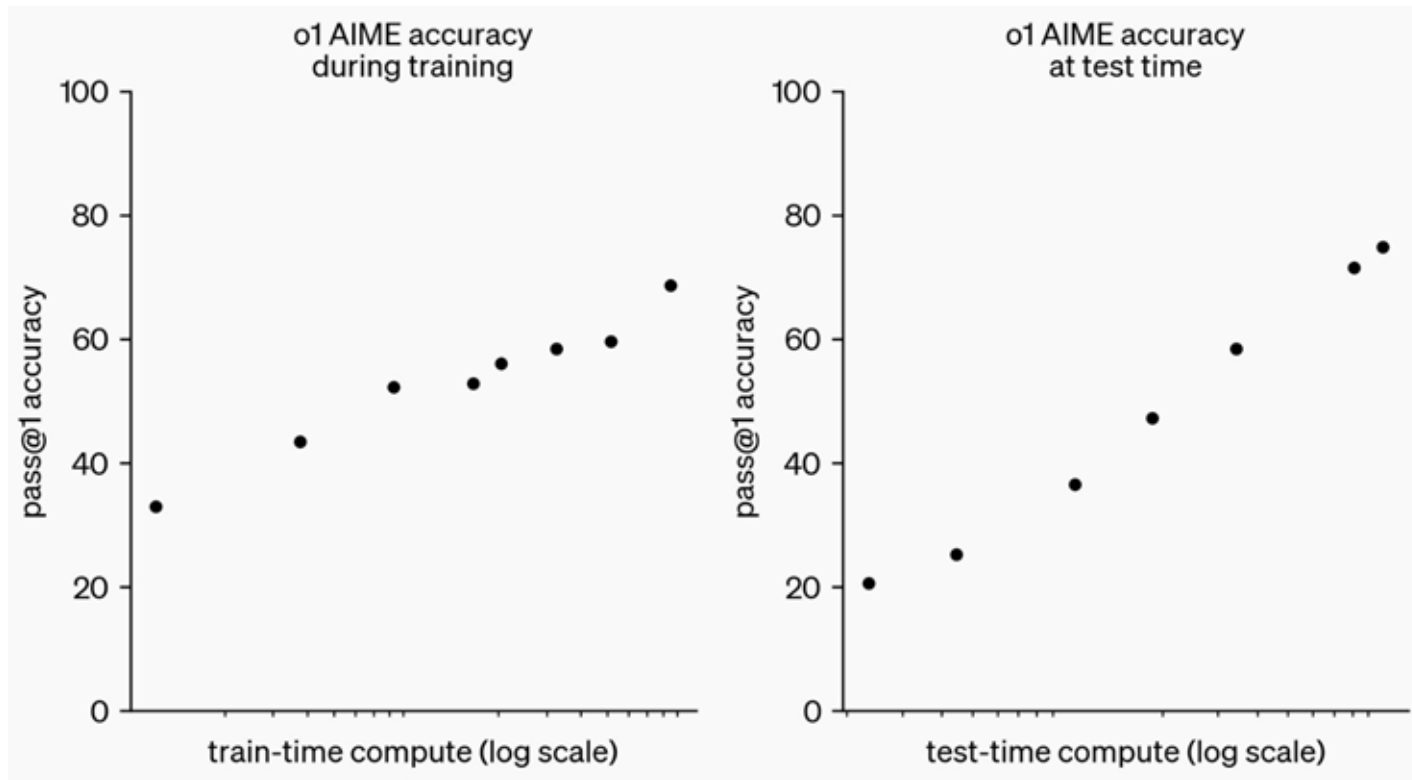
The model was trained with a correctness reward only. Re-checking its own steps appeared during RL training and was not written into the prompt.

2024-2025 RL-trained reasoning



GRPO removes the value network used in PPO and estimates advantages by comparing rewards across a group of rollouts for the same prompt.

2024-2025 RL-trained reasoning



2025 Multi-turn Agentic Reasoning

RL extends from single-response reasoning to complex multi-turn interaction

Curious is a women's fragrance by a singer born in what city and state?

<think> Find the singer behind "Curious", then her birthplace.

<search> Curious fragrance information

<information> Curious is a women's fragrance by Britney Spears ...

<search> Britney Spears birthplace

<information> ... born in McComb, Mississippi ...

<answer> McComb, Mississippi ✓ **reward = 1**

2025 Multi-turn Agentic Reasoning

These details will help ensure the report is as precise and actionable as possible.

Penetration as a percentage, and look at overall usage. Make your best assumptions on the rest!



Got it! I will research the iOS and Android adoption rates, the percentage of people interested in learning another language, and the change in mobile penetration (as a percentage of the population) over the past five years. This will be analyzed for the top 10 developed and top 10 developing countries by GDP, using the best available benchmarks.

The findings will be compiled into a structured report with a table summarizing key metrics, along with recommendations on the most promising markets for a new translation app from ChatGPT, focusing on markets where ChatGPT could expand further.

I will update you once the report is ready.

Searched for top 10 developing countr...

1 source

Message ChatGPT

+

 Search

 Deep research

...



ChatGPT can make mistakes. Check important info.

country

 I'm looking into recent surveys and reports on smartphone penetration and OS adoption to understand digital trends and language interest globally.

I'm piecing together details on Android and iOS market share trends over the past five years, focusing on developing and developed countries, to understand regional differences and shifts.

 Searched for top 10 countries by GDP 2023

 I'm mapping out the largest advanced economies, including the US, Japan, Germany, UK, France, Italy, Canada, South Korea, Australia, and Spain.

 Read en.wikipedia.org

 I'm thinking through a Wikipedia page and a table for all countries. Considering a quicker, known method to identify the top 10 developed versus developing countries, or if a pre-listed "top 10 developing countries by GDP" is available.

 Searched for top 10 developing countries by GDP

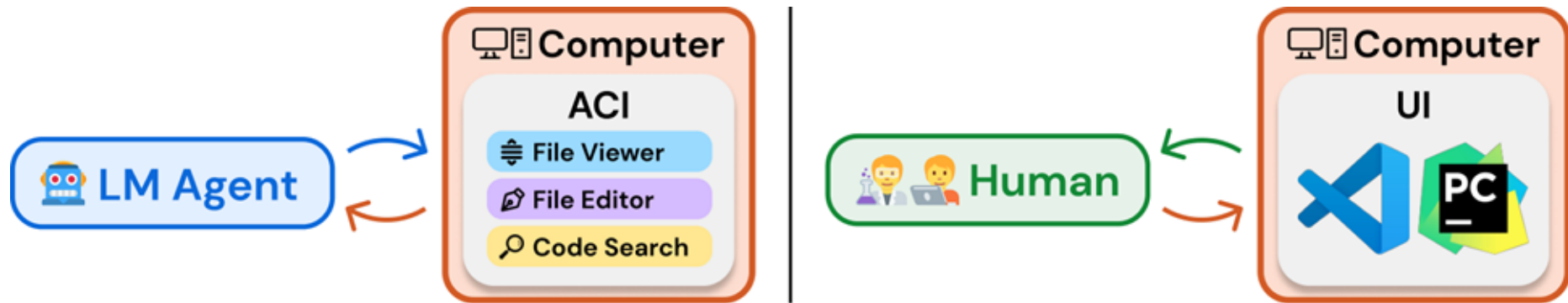
 Thinking...

Employee preview  

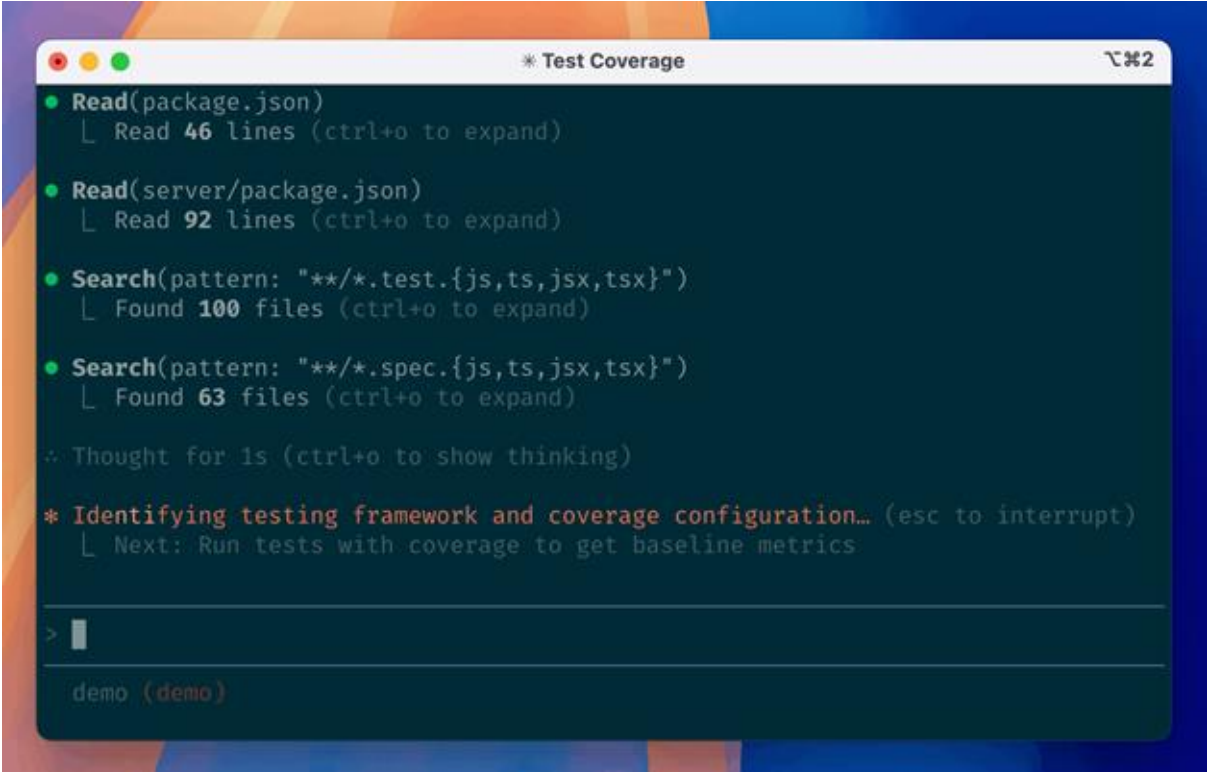
Introducing Deep Research, OpenAI, February 2025.

2024-2026 Long-horizon model-harness agent systems

The harness enables agents to sustain progress on long-horizon tasks through context management, memory, tools, and orchestration.



2024-2026 Long-horizon model-harness agent systems



```
* Test Coverage Ƶ#2

• Read(package.json)
  | Read 46 lines (ctrl+o to expand)

• Read(server/package.json)
  | Read 92 lines (ctrl+o to expand)

• Search(pattern: "**/*.test.{js,ts,jsx,tsx}")
  | Found 100 files (ctrl+o to expand)

• Search(pattern: "**/*.spec.{js,ts,jsx,tsx}")
  | Found 63 files (ctrl+o to expand)

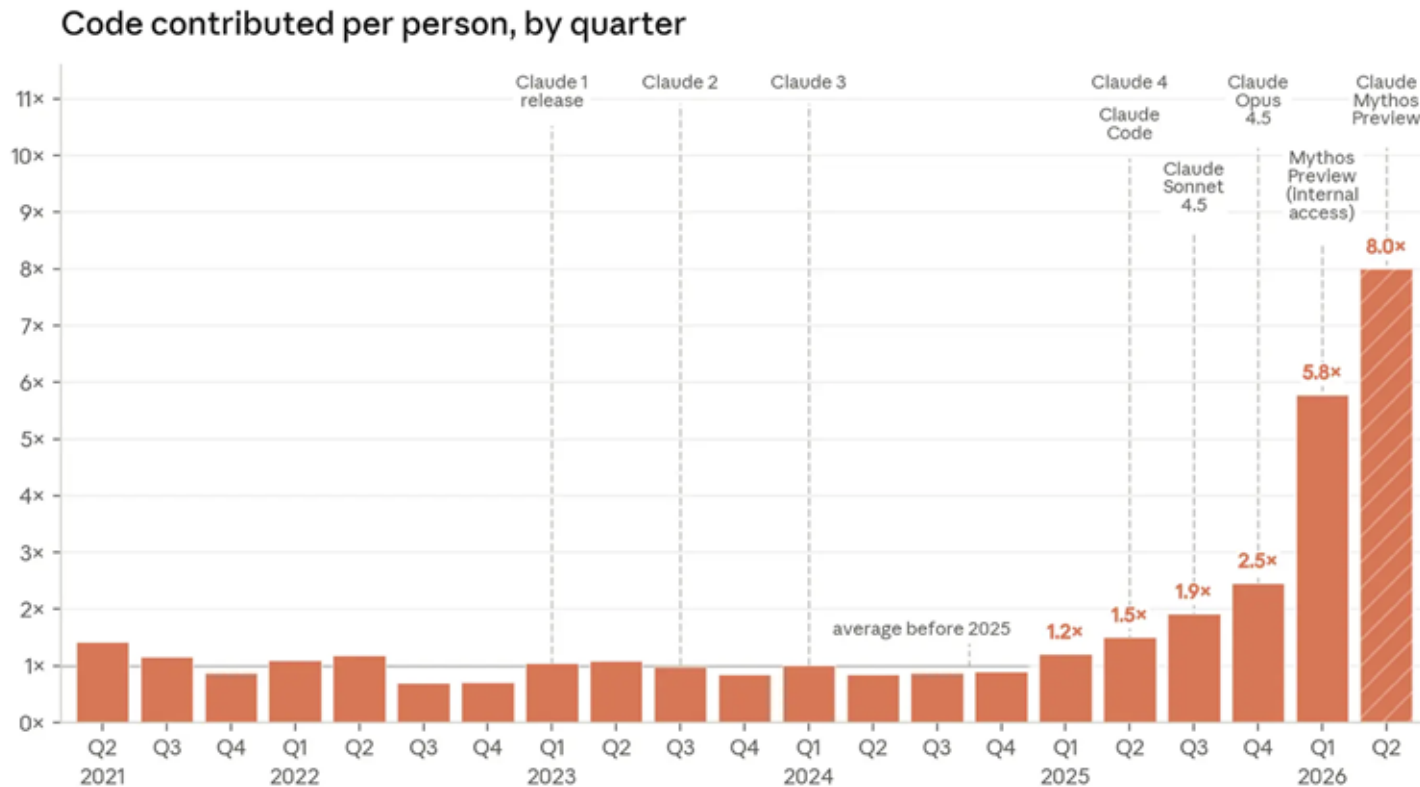
✧ Thought for 1s (ctrl+o to show thinking)

* Identifying testing framework and coverage configuration... (esc to interrupt)
  | Next: Run tests with coverage to get baseline metrics

> |

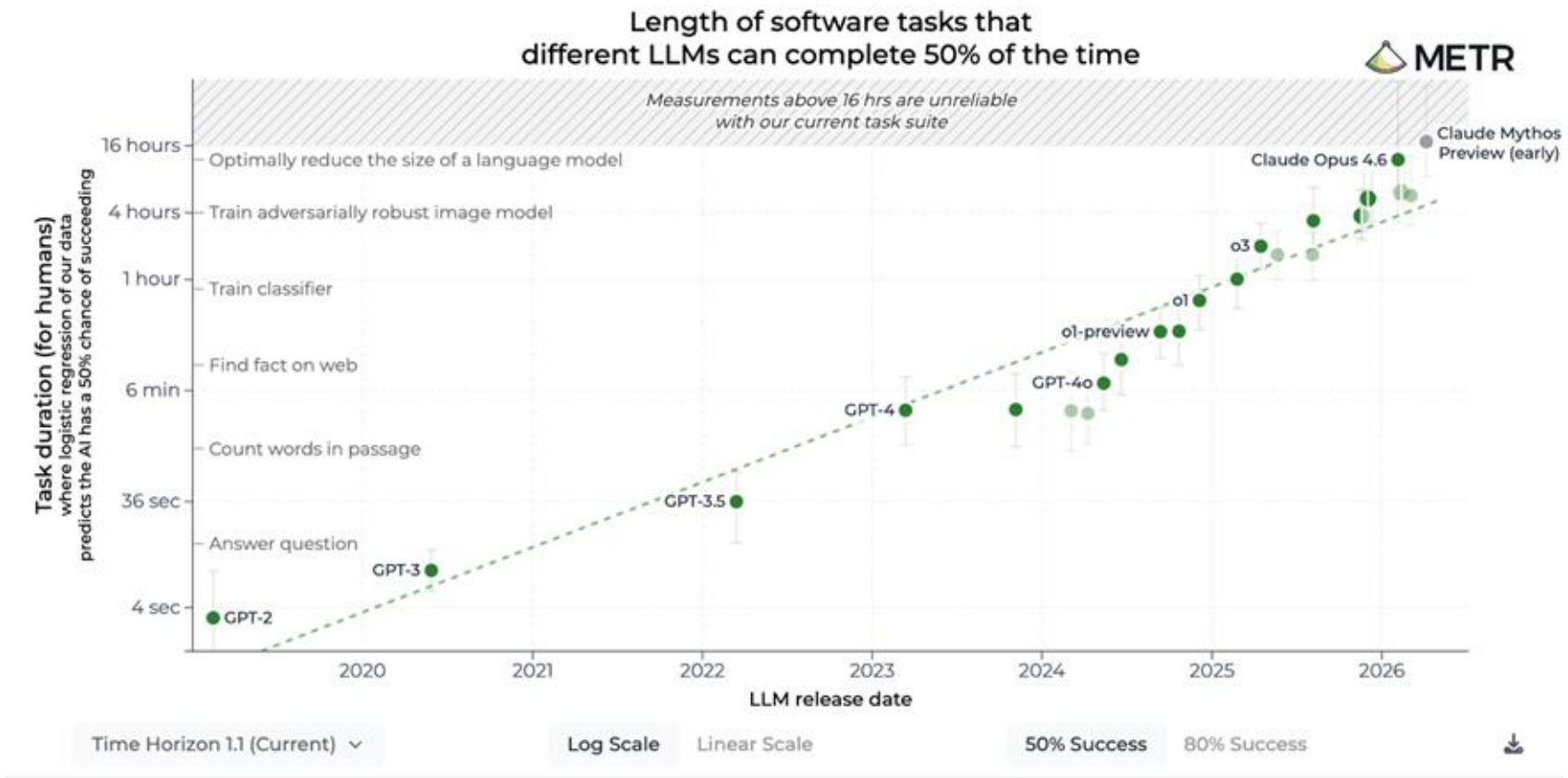
demo (demo)
```

2024-2026 Long-horizon model-harness agent systems

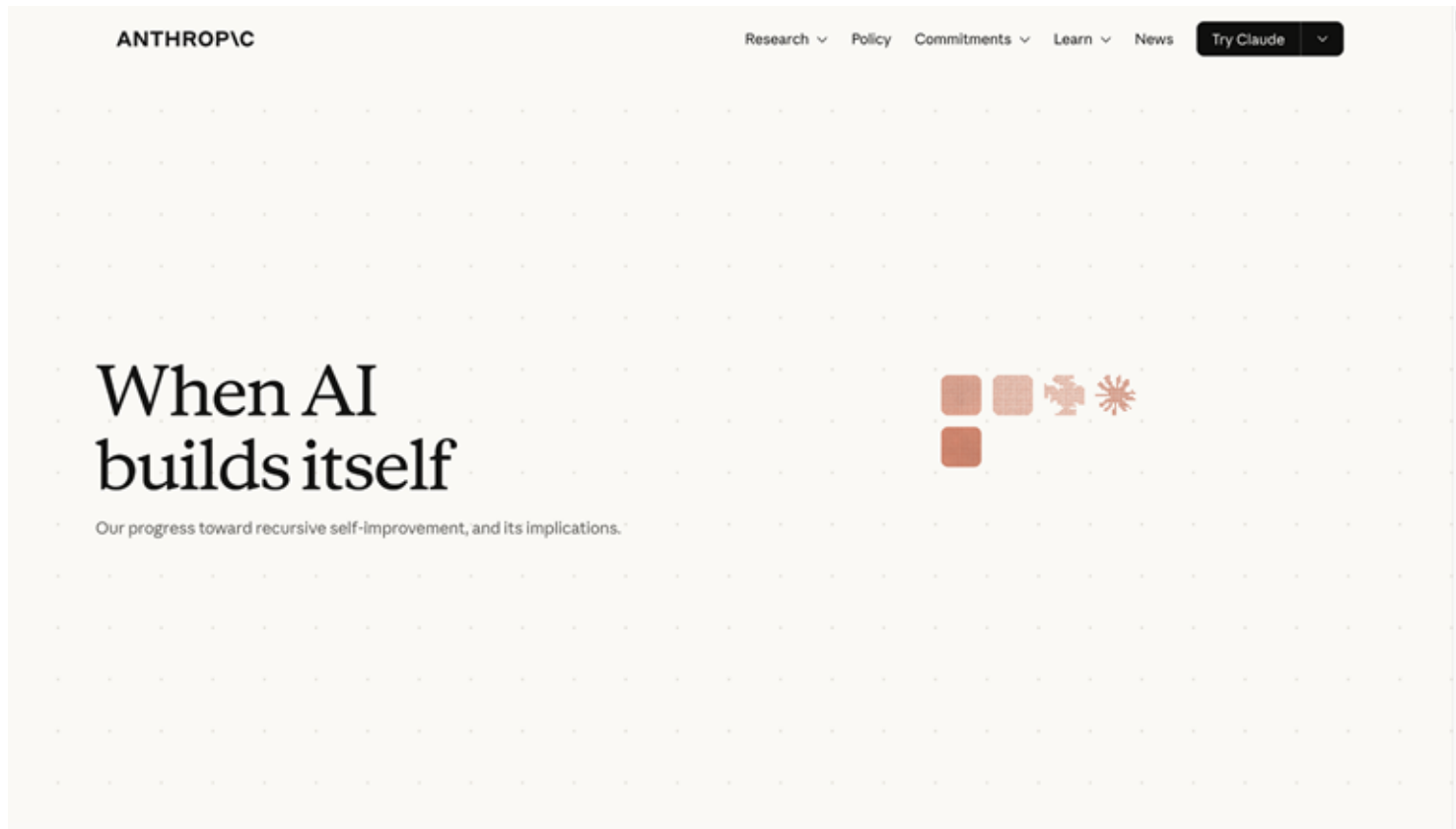


Code written per contributor at Anthropic has increased eightfold this year, and Claude writes approximately 80% of the code merged at Anthropic.

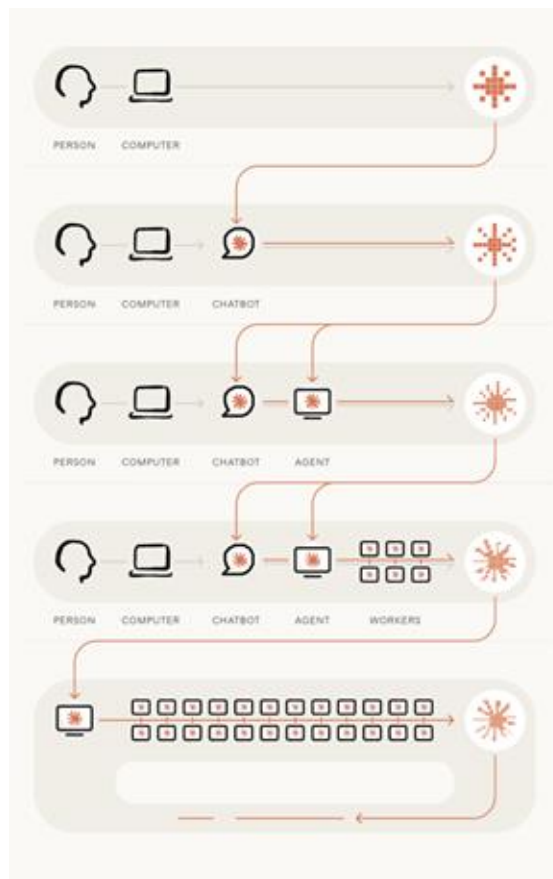
2024-2026 Long-horizon model-harness agent systems



2026 Automated Improvement Loops?



2026 Automated Improvement Loops?



Building the first Claude

In the early days, work at Anthropic looked like work at any other tech company: people writing code and docs on laptops.

2023-2023

Chatbots

People used early chatbots to help with parts of the process, like generating short code snippets and copying the output into text editors.

2025-2026

Coding agents

As the agents became more capable, they were able to write and edit code on their own, sometimes entire files.

TODAY

Autonomous agents

Agents can now run code themselves and delegate hours of work to other agents.

20XX?

Closing the loop

In the future, agents could become capable enough to build and train models themselves. If this happens, future versions of Claude could be continuously improved by Claude itself.

2026 Automated Improvement Loops?

OpenAI

ResearchProductsBusinessDevelopersCompanyFoundation

Q

Log in

Try ChatGPT

September 6, 2026

ResearchPublicationSafety

Research acceleration: The view inside OpenAI

▶ Listen to article

13:39

🔗 Share

1. Coding agents are reshaping daily work for OpenAI researchers

2. Researchers are writing more code and running more experiments

3. The work researchers use agents for is changing

4. Pacing model development

5. The path ahead

Appendix: Our methods for this post

For AGI to benefit all of humanity, we believe it must be democratically governed. This can only happen through an informed public debate about the capabilities, risks and safeguards of highly capable AI systems. People everywhere need to understand the likely future trajectory of frontier AI, so they can have a meaningful voice in how it develops.

Transparency about specific risks, incidents and safeguards is necessary, but not sufficient. We believe the public also needs to understand how the most capable systems are developing, and how they are driving research progress, inside of frontier labs.

We aim to safely build an automated AI researcher that can work under human supervision to further progress on deep learning and alignment, enabling iterative improvements. According to our measurements, we have now reached the goal,

History of self-evolving AI,
and why is it blowing up now?

1965 Intelligence Explosion

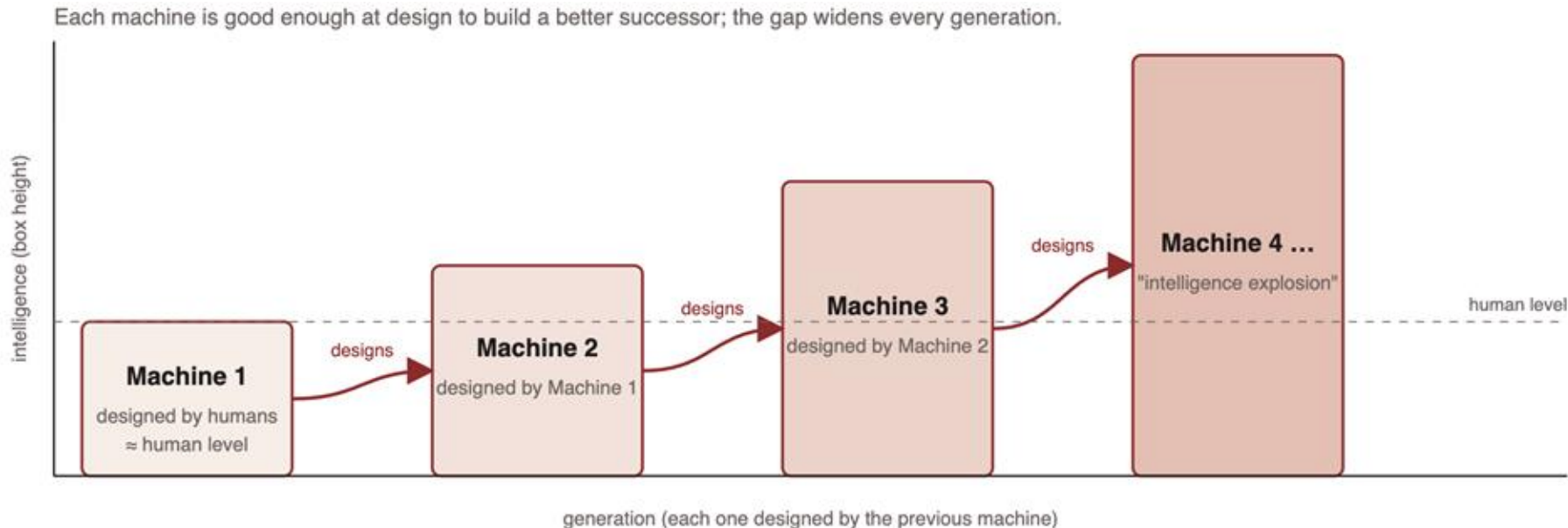
Irving John Good was a British mathematician who worked as a cryptologist at Bletchley Park alongside Alan Turing.

After the Second World War, Good continued working with Turing on computer design and Bayesian statistics at the University of Manchester.

In 1965, he published “Speculations Concerning the First Ultraintelligent Machine,” in which he proposed the idea of **an intelligence explosion driven by machines recursively improving their own design.**

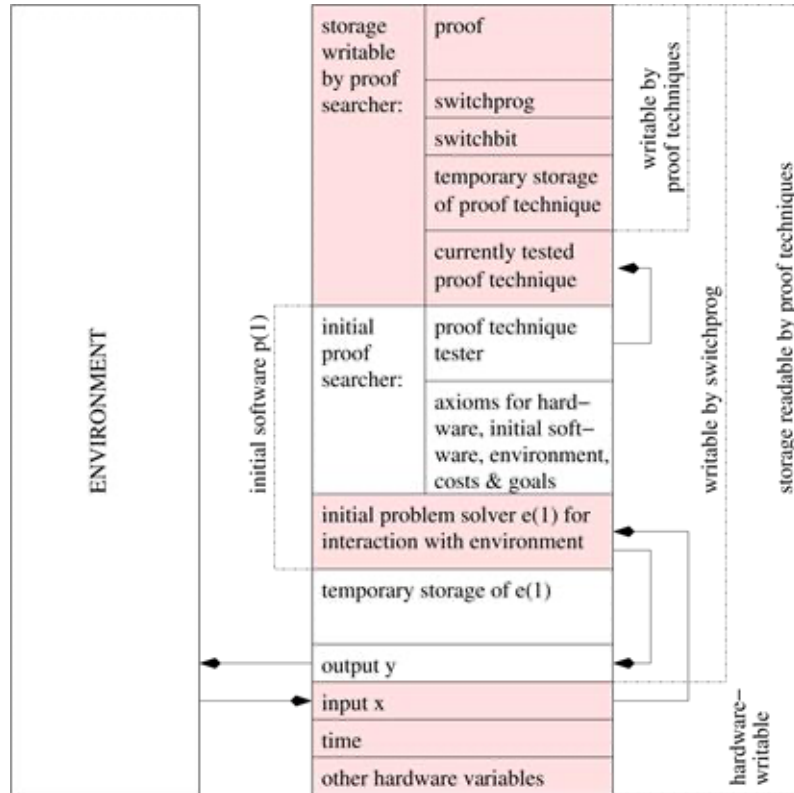


1965 Intelligence Explosion



1. A machine has the intellectual ability to design machines.
2. It uses that ability to design a more capable successor.
3. That successor is also better at machine design.
4. It can therefore contribute to another round of improvement.

2003 Gödel Machines



Jürgen Schmidhuber

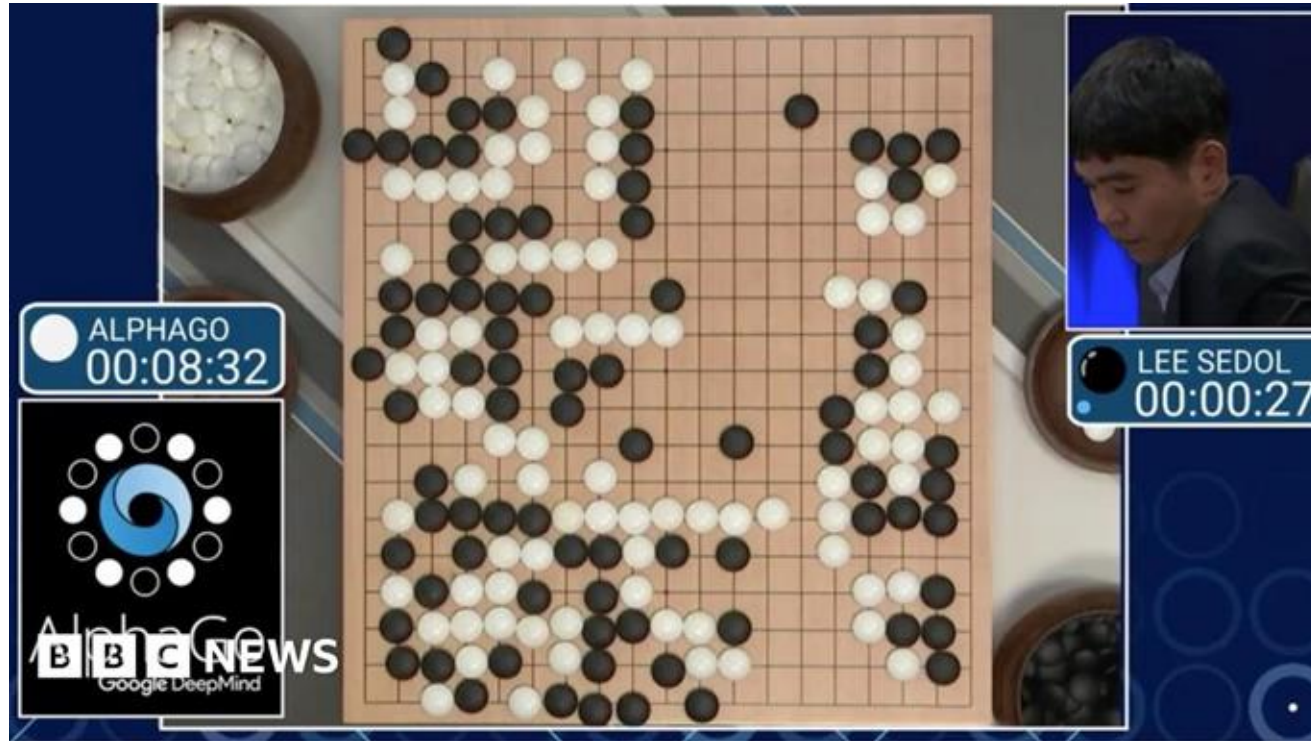
Gödel Machines: Self-Referential Universal Problem Solvers Making Provably Optimal Self-Improvements, J. Schmidhuber (IDSIA), 2003.

2003 Godel Machines

```
loop forever:
    solve problems in the environment          # earns utility
    propose switchprog                        # a rewrite of own code
    search for a proof that
        utility( run switchprog ) > utility( keep going )
    if proof found: execute switchprog        # may rewrite the prover too
```

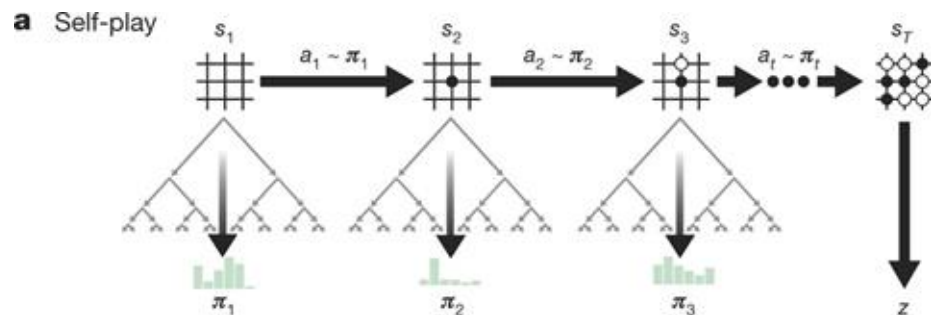
A self-referential system modifies its own code when it can prove that the modification is beneficial.

2017 AlphaGo Zero - Self-Play At Scale

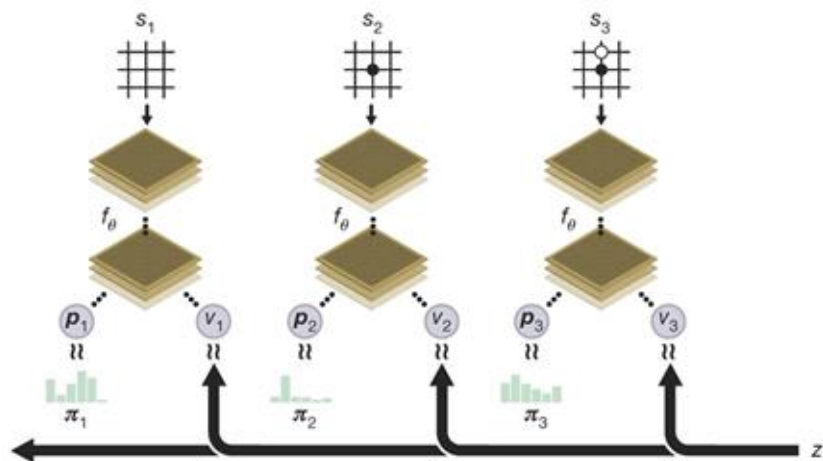


AlphaGo is first trained to imitate expert gameplay trajectories before reinforcement learning.

2017 AlphaGo Zero - Self-Play At Scale



b Neural network training



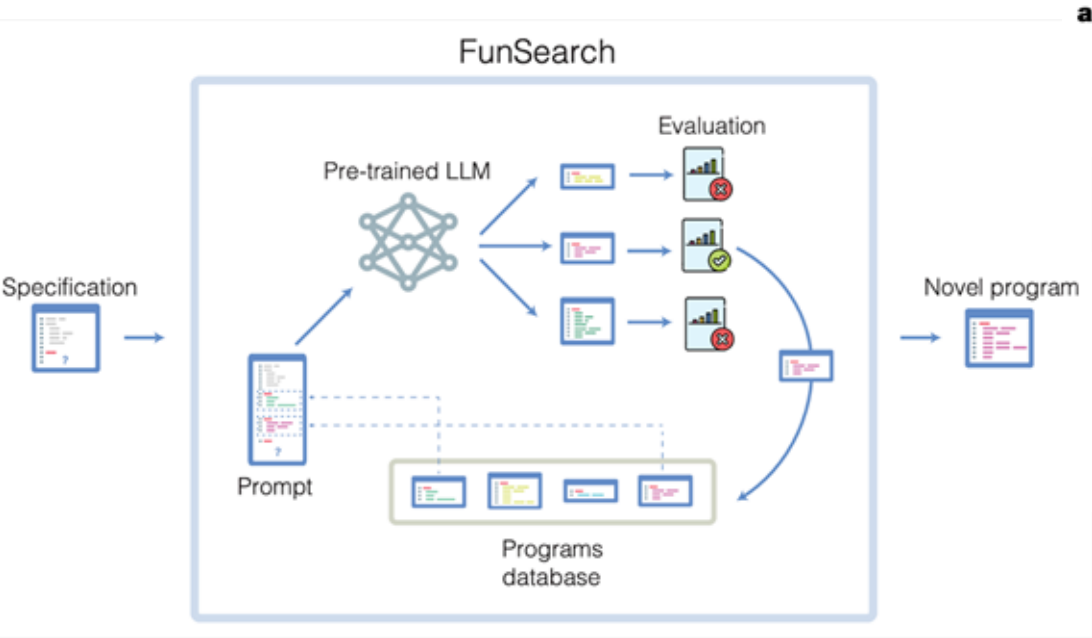
David Silver

2017 AlphaGo Zero - Self-Play At Scale



AlphaGo Zero improves through a repeated cycle of self-play without human gameplay data.

2023 LLM-guided discovery



a

n	3	4	5	6	7	8
Best known	9	20	45	112	236	496
FunSearch	9	20	45	112	236	512

Cap Set Problem

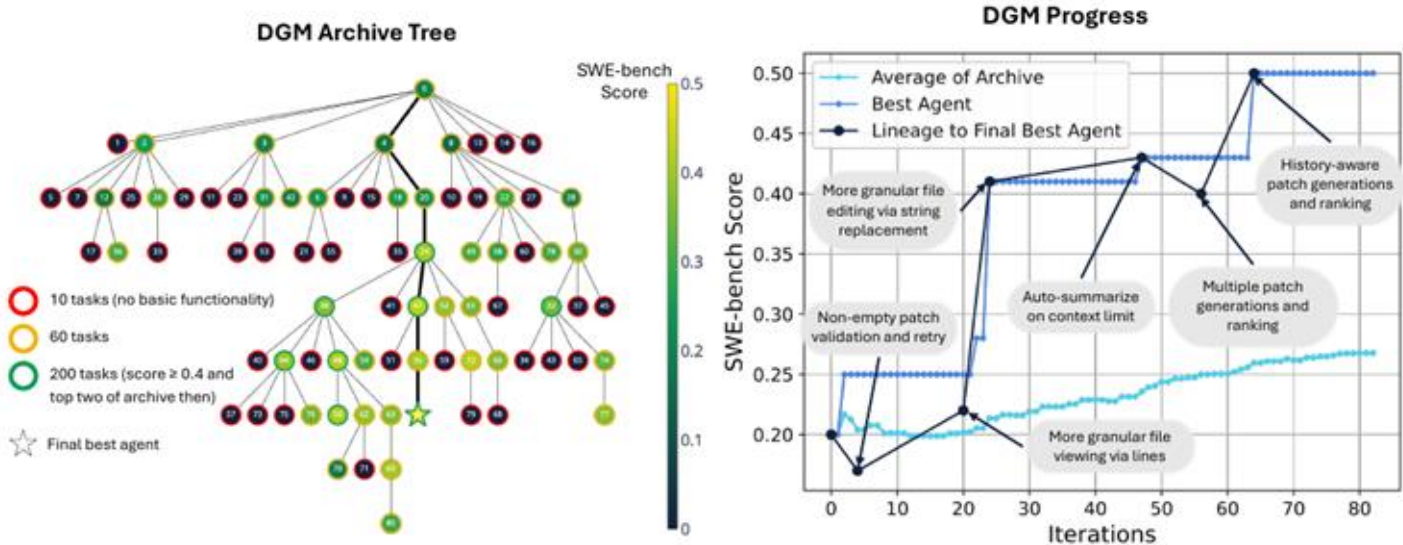
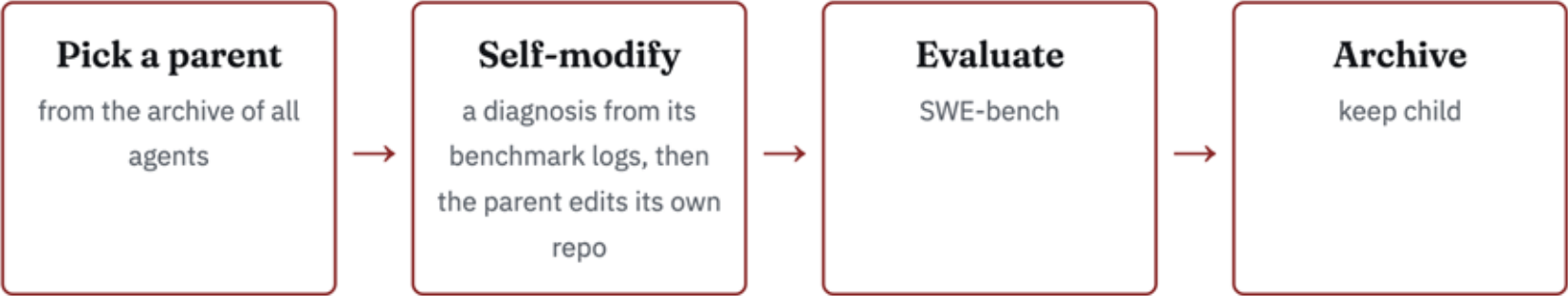
	OR1	OR2	OR3	OR4	Weibull 5k	Weibull 10k	Weibull 100k
First fit	6.42%	6.45%	5.74%	5.23%	4.23%	4.20%	4.00%
Best fit	5.81%	6.06%	5.37%	4.94%	3.98%	3.90%	3.79%
FunSearch	5.30%	4.19%	3.11%	2.47%	0.68%	0.32%	0.03%

Fraction of excess bins (lower is better) for various bin packing heuristics on the OR and Weibull datasets. FunSearch outperforms first fit and best fit across problems and instance sizes.

Online Bin Packing

FunSearch combines LLM-generated code with automated evaluation in an evolutionary loop to discover new cap set constructions and improved bin-packing heuristics.

2025 Self-Modifying Agent

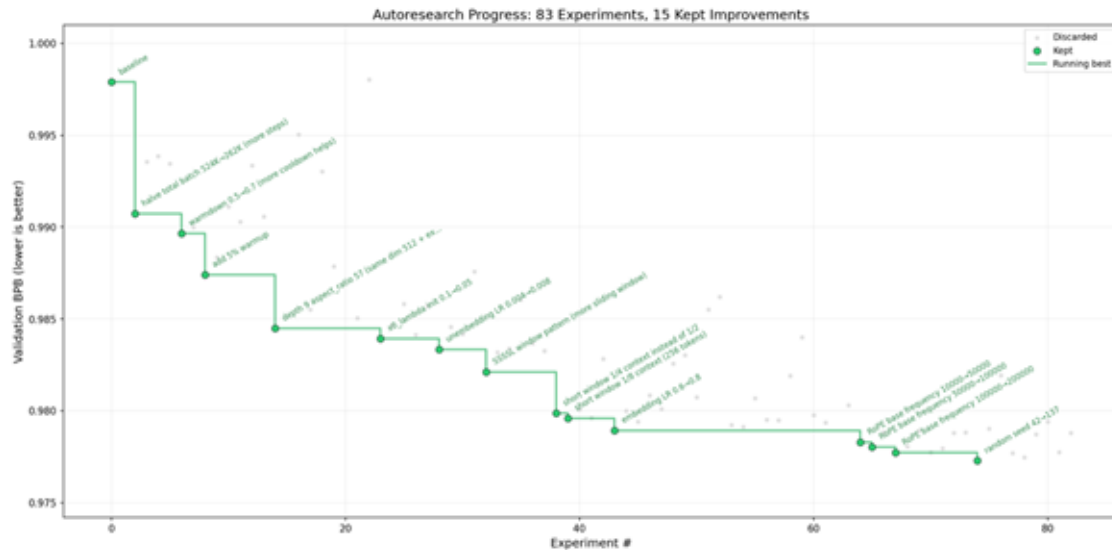


2026 Automated AI research

PROGRAM.MD, "THE EXPERIMENT LOOP"

LOOP FOREVER:

1. Look at the git state
2. Tune `train.py` with an experimental idea
3. git commit
4. Run: `uv run train.py > run.log`
5. `grep "^val_bpb:" run.log`
6. If empty, the run crashed; read the trace
7. Record the results in the tsv
8. If val_bpb improved, keep the commit
9. If equal or worse, git reset



An AI agent follows program.md to propose ideas, edit code, and run experiments that improve language model pretraining under a fixed compute budget (5 minutes per experiment, ~100 experiments one night)

Today's Landscape of Self-Evolving AI

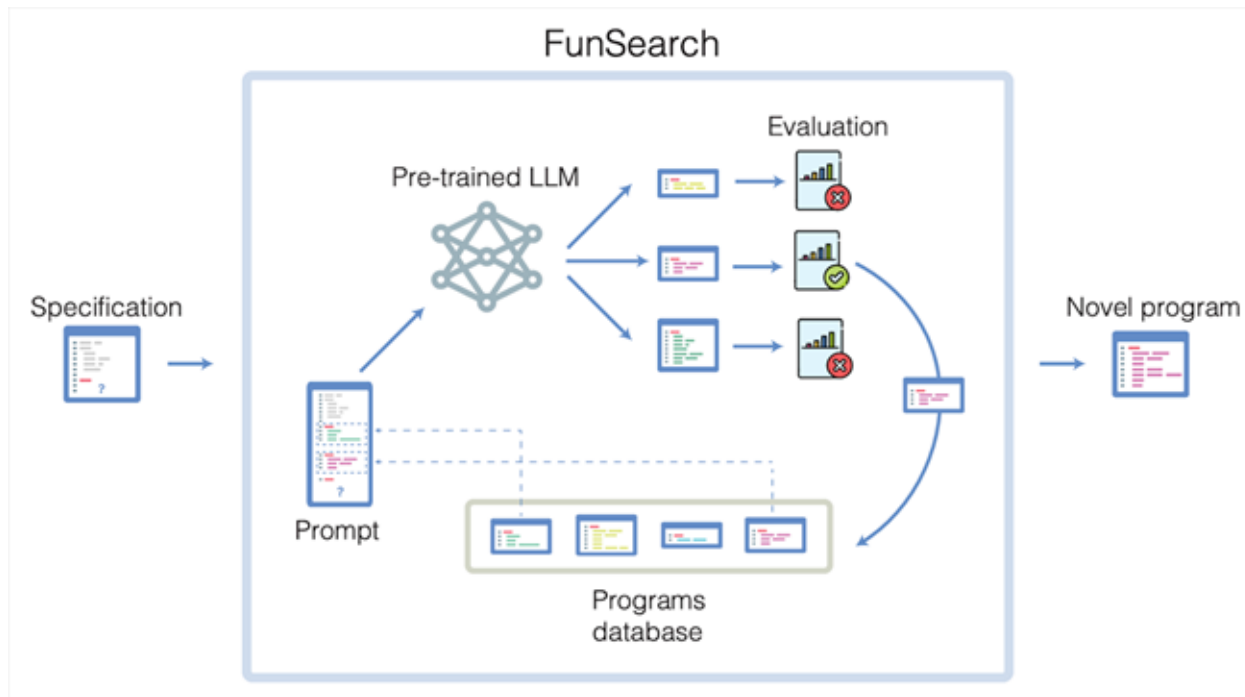
Perspective 1: What is being evolved?

Contexts & artifacts

prompts · answers · programs · archives

FunSearch · DSPy · PromptBreeder

Perspective 1: What is being evolved?



In FunSearch, math programs are
being evolved

Perspective 1: What is being evolved?

Contexts & artifacts

prompts · answers · programs · archives

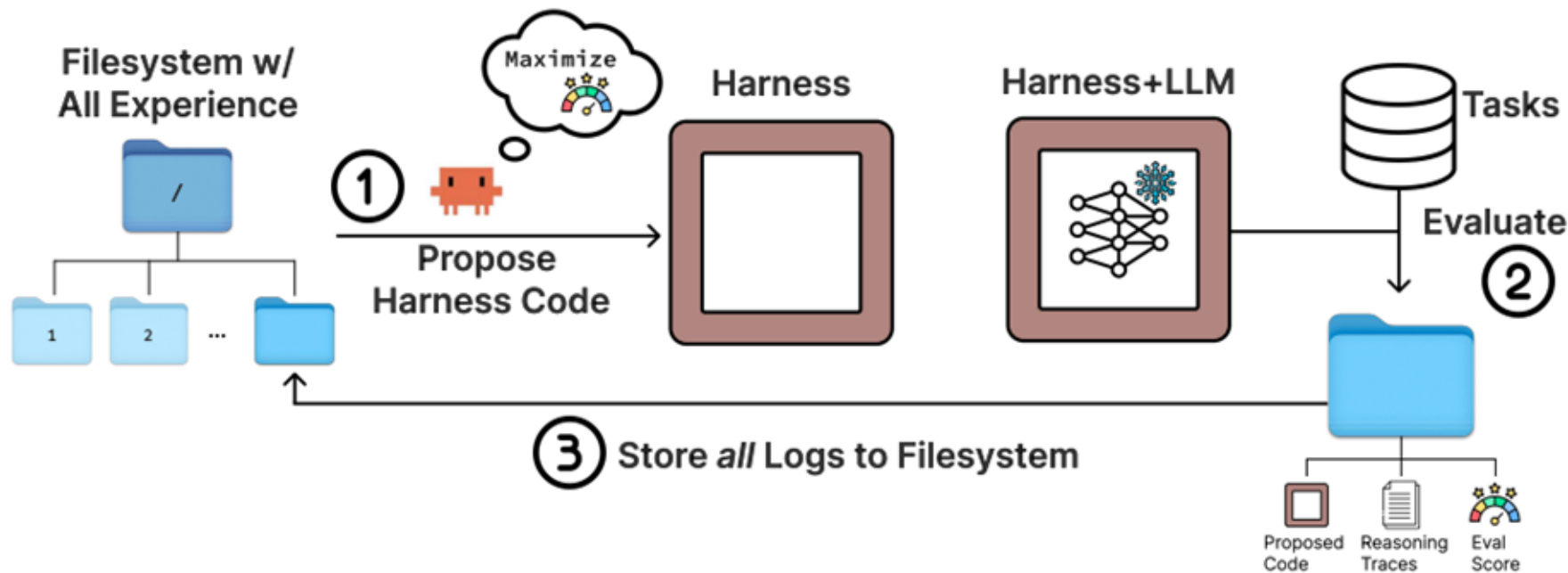
FunSearch · DSPy · PromptBreeder

Memory & harness

memory · skills · tools · scaffold code

Voyager · DGM · Meta-Harness

Perspective 1: What is being evolved?



In Meta-Harness, the agent harness is being evolved

Perspective 1: What is being evolved?

Contexts & artifacts

prompts · answers · programs · archives

FunSearch · DSPy · PromptBreeder

Memory & harness

memory · skills · tools · scaffold code

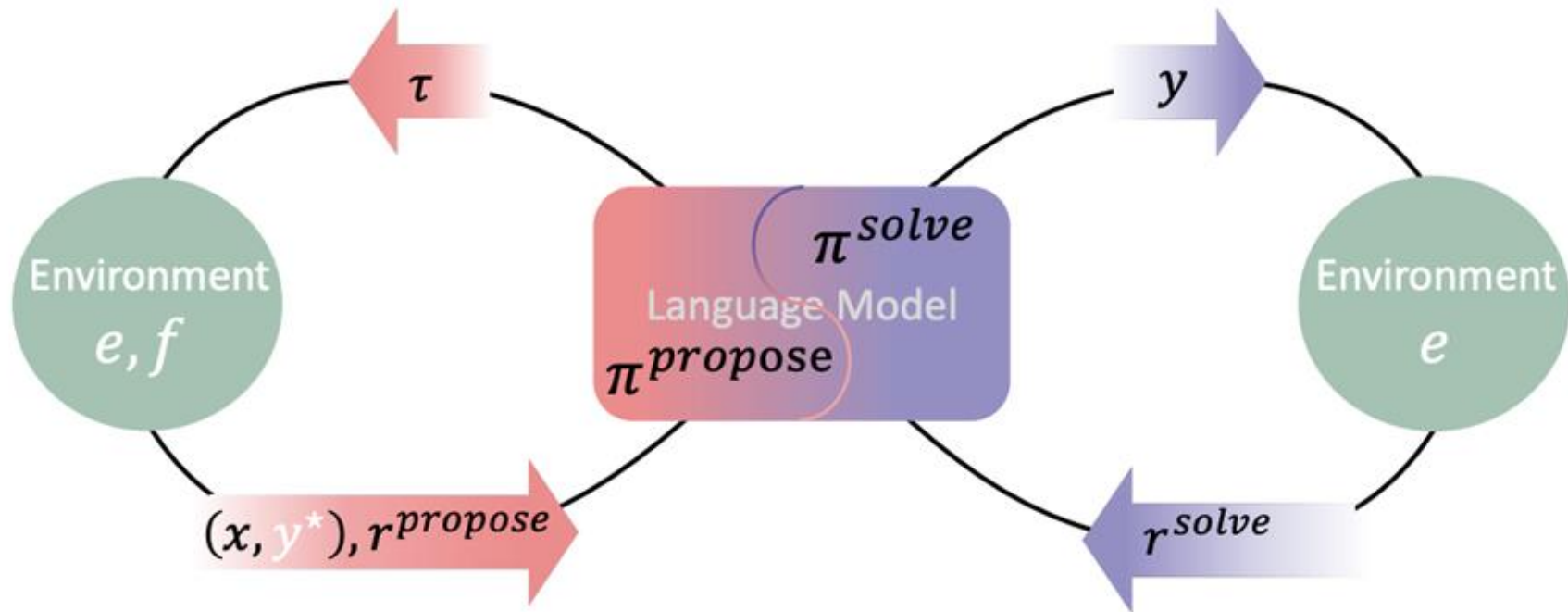
Voyager · DGM · Meta-Harness

Model weights

weights · adapters

STaR · DeepSeek-R1 · Absolute Zero

Perspective 1: What is being evolved?



In Absolute Zero, model weights are being evolved

Perspective 1: What is being evolved?

Contexts & artifacts

prompts · answers · programs · archives

FunSearch · DSPy · PromptBreeder

Memory & harness

memory · skills · tools · scaffold code

Voyager · DGM · Meta-Harness

Model weights

weights · adapters

STaR · DeepSeek-R1 · Absolute Zero

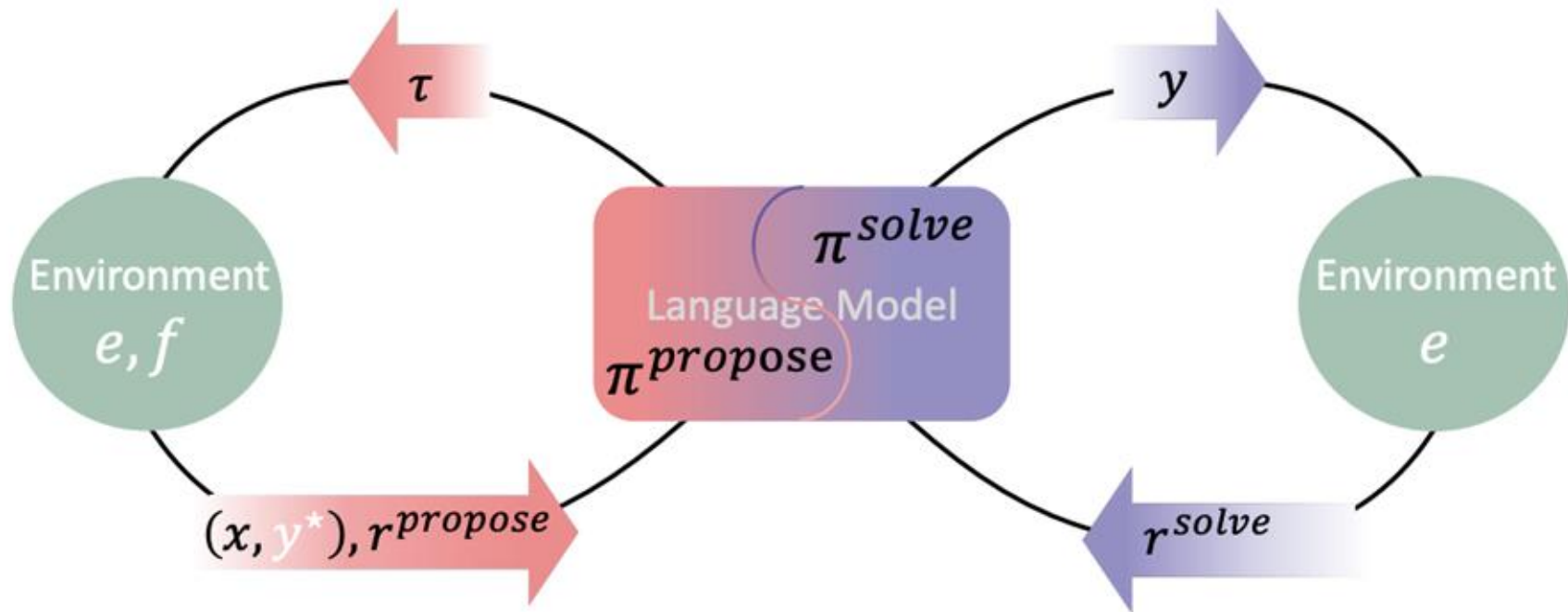
Environment & tasks

tasks · curricula · environments · verifiers

(improvement mechanisms)

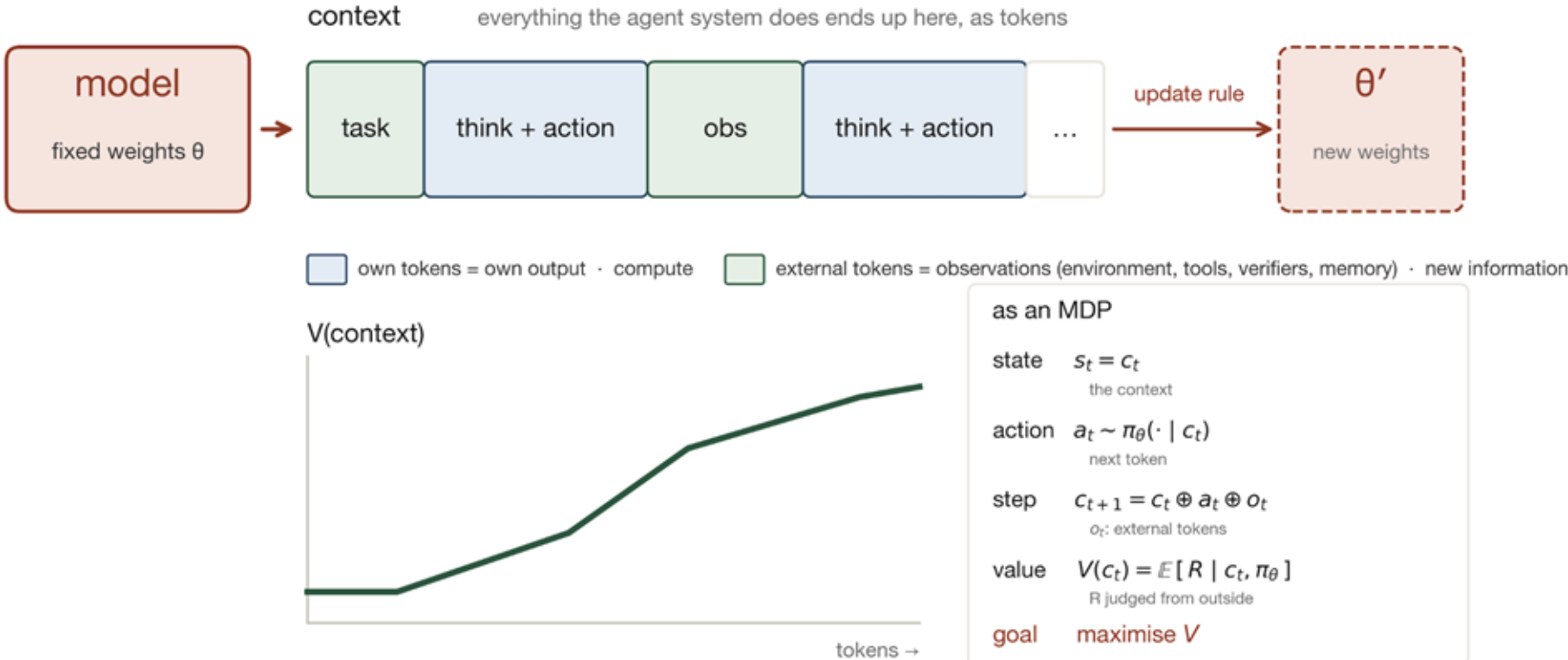
SPADE · OMNI-EPIC · Self-Challenging

Perspective 1: What is being evolved?



In Absolute Zero, tasks are also evolved

Perspective 2: How is it being evolved? A Model View



Self-improvement: keep adding tokens (or changing θ) so that V keeps rising.

Perspective 2: How is it being evolved? An Agent View



Perspective 2: How is it being evolved? An Agent View

Step 1

Collect demonstration data, and train a supervised policy.

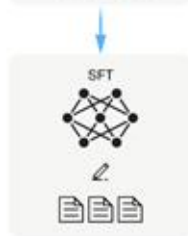
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

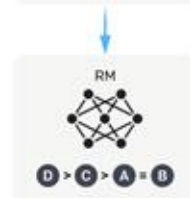
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



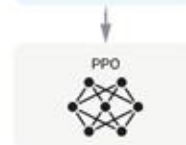
Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.



The policy generates an output.

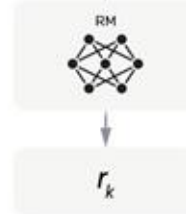


Once upon a time...

The reward model calculates a reward for the output.

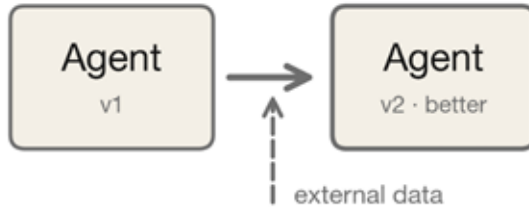


The reward is used to update the policy using PPO.



Perspective 2: How is it being evolved? An Agent View

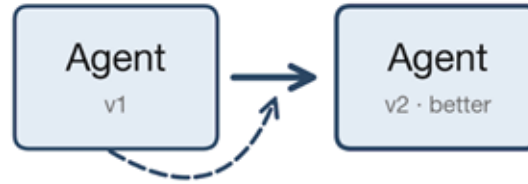
IMPROVED BY OTHERS



external data,
a fixed update rule

RLHF · GRPO

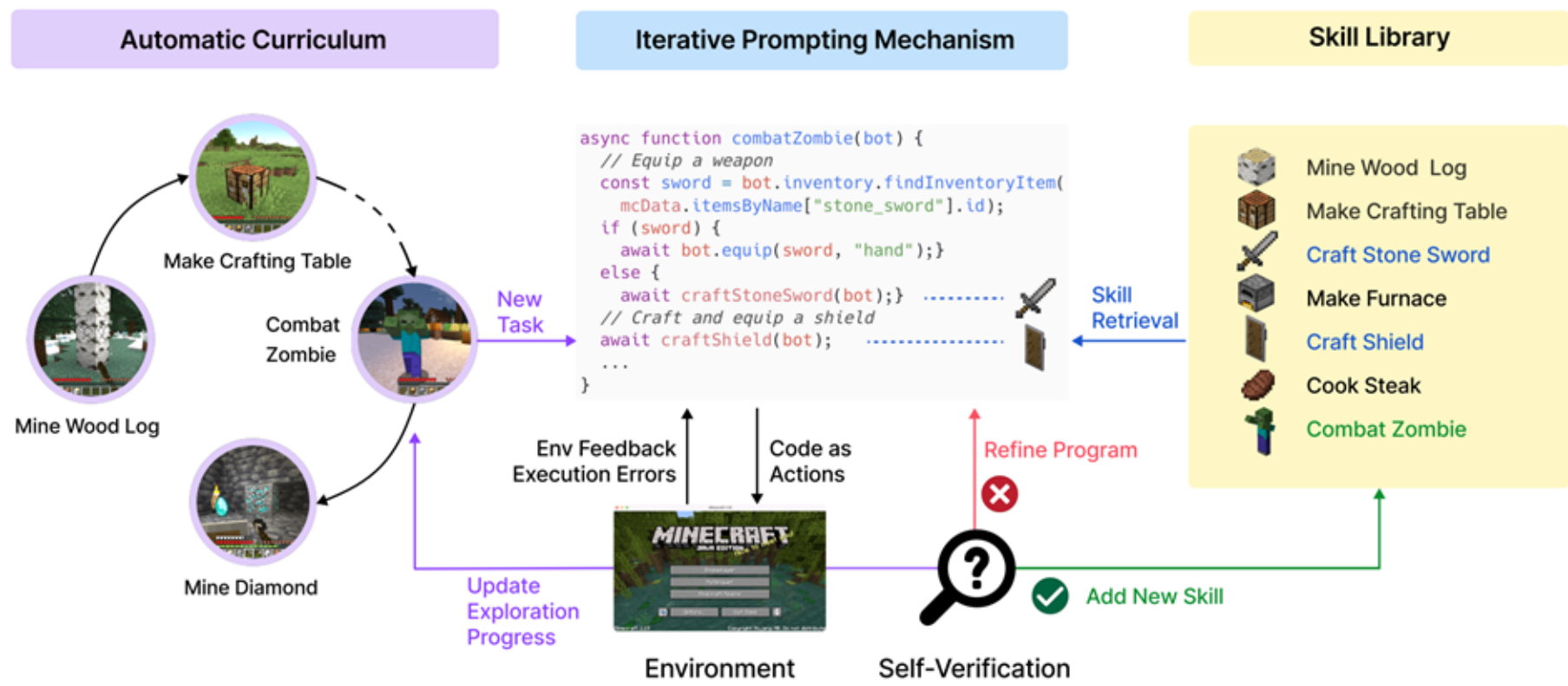
SELF-EVOLVING



its own experience,
a fixed update rule

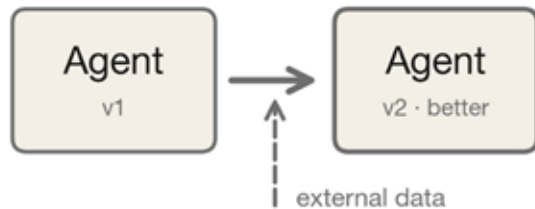
STaR · Voyager · AlphaEvolve

Perspective 2: How is it being evolved? An Agent View



Perspective 2: How is it being evolved? An Agent View

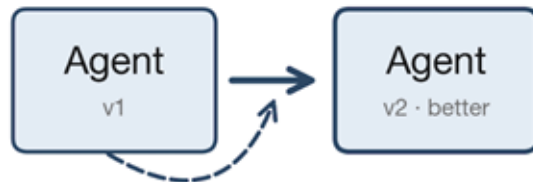
IMPROVED BY OTHERS



external data,
a fixed update rule

RLHF · GRPO

SELF-EVOLVING

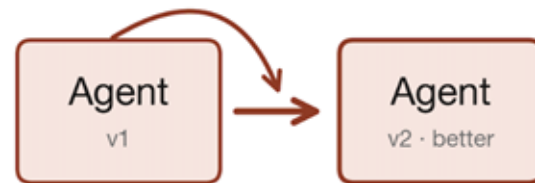


its own experience,
a fixed update rule

STaR · Voyager · AlphaEvolve

SELF-REFERENTIAL

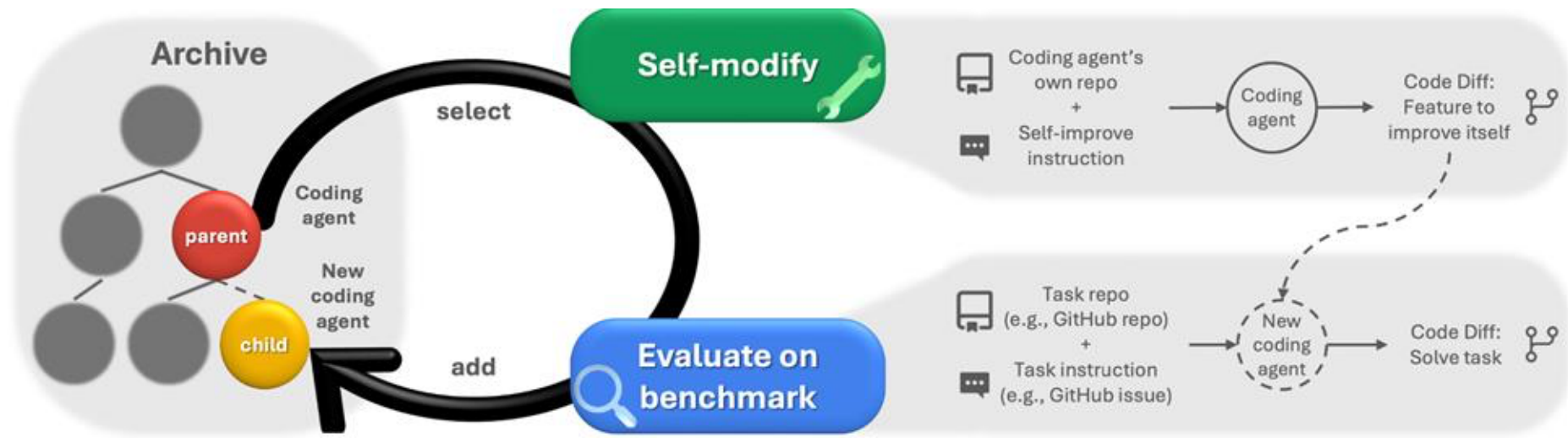
= recursive self-improvement (structural sense)



its own experience,
the agent rewrites itself

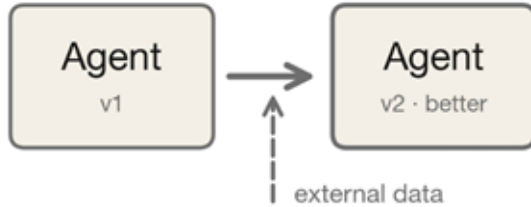
DGM · PromptBreeder · HyperAgents

Perspective 2: How is it being evolved? An Agent View



Perspective 2: How is it being evolved? An Agent View

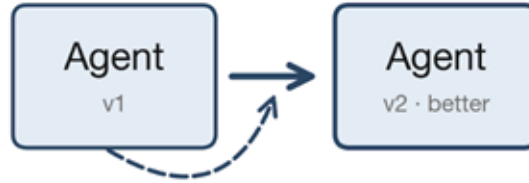
IMPROVED BY OTHERS



external data,
a fixed update rule

RLHF · GRPO

SELF-EVOLVING

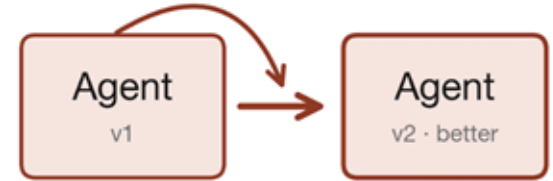


its own experience,
a fixed update rule

STaR · Voyager · AlphaEvolve

SELF-REFERENTIAL

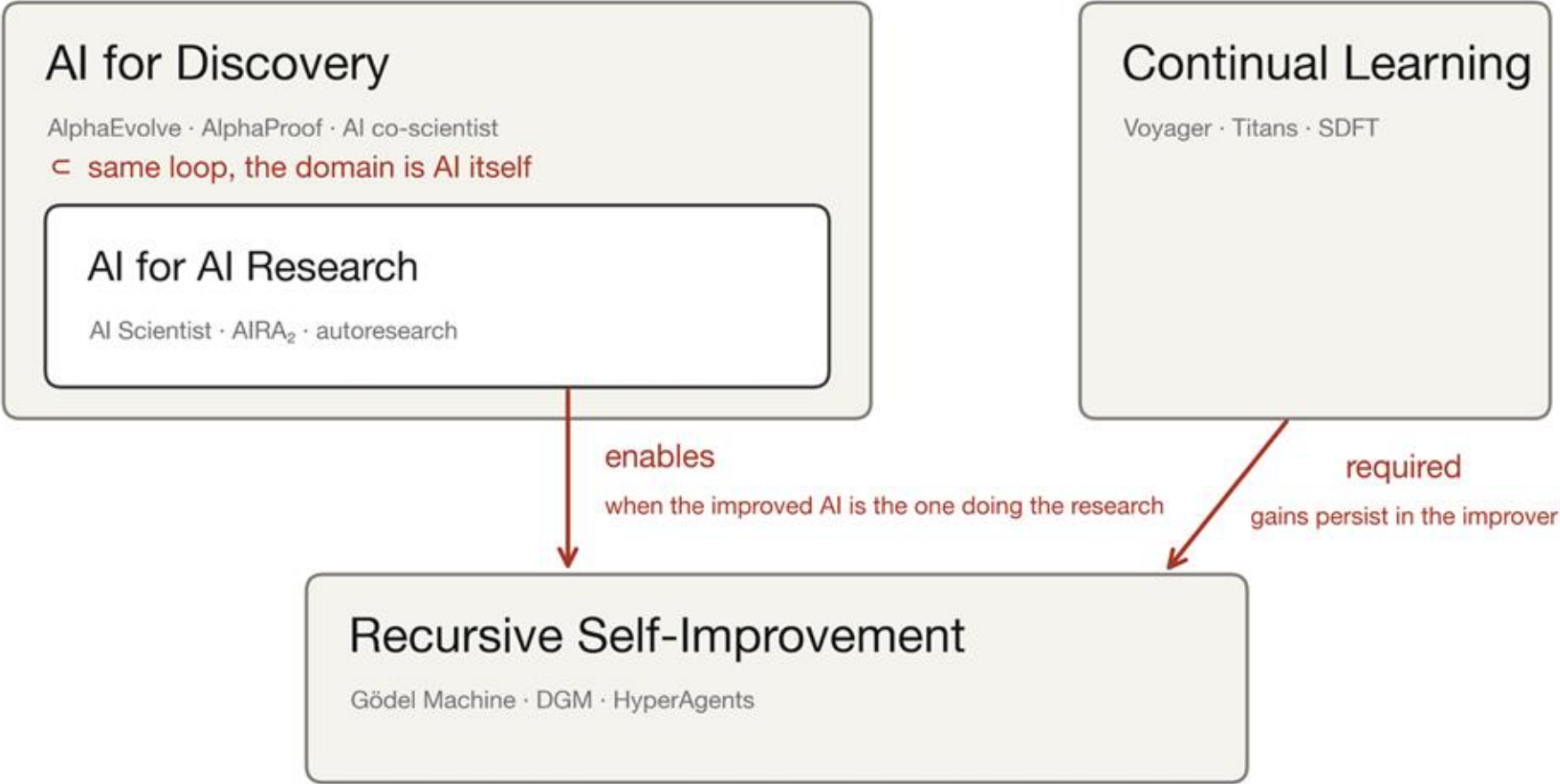
= recursive self-improvement (structural sense)



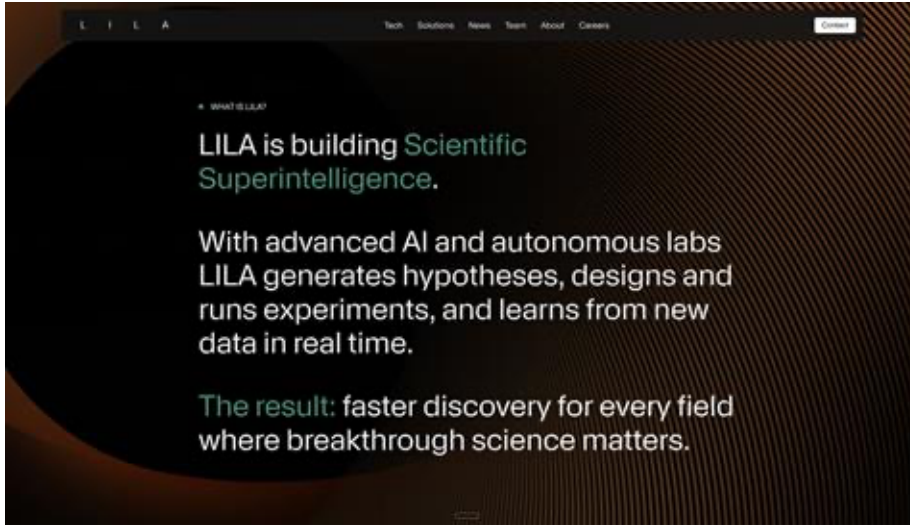
its own experience,
the agent rewrites itself

DGM · PromptBreeder · HyperAgents

Perspective 3: Relationships between big concepts



AI for Discovery



LILA Sciences

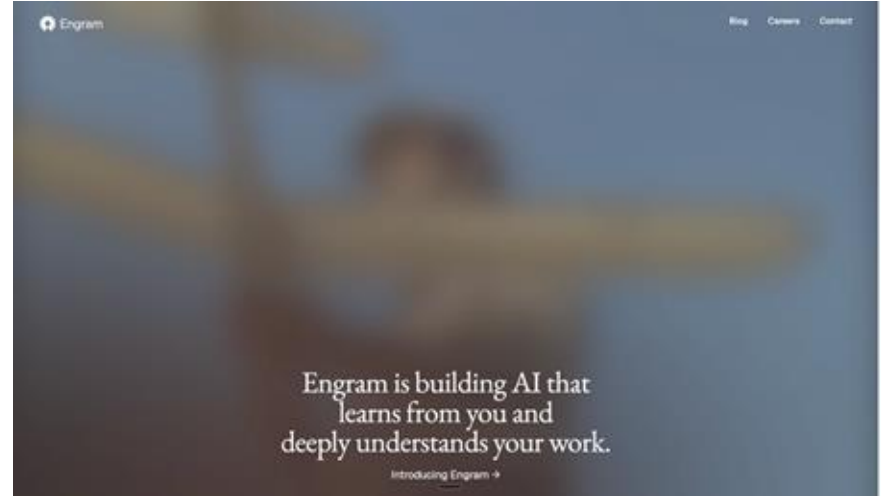


Discovery Loop

Continual Learning

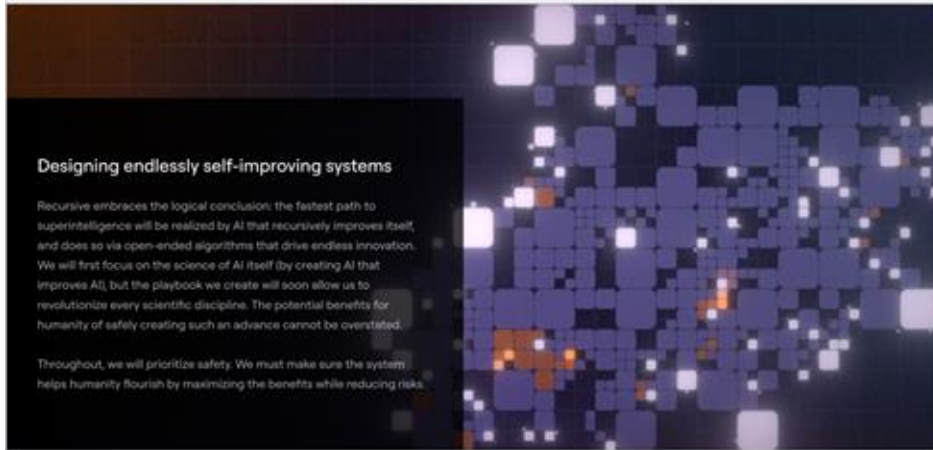


Oak Lab



Engram

AI for AI Research -> RSI



Recursive Superintelligence



Mirendil

Research Ideas and Tasks

Can LLM learn taste and make genuine innovations?

Goal: move beyond generating plausible variations of existing ideas toward identifying valuable directions and producing genuinely novel discoveries.

1. Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers, Si et al., 2024.
2. HindSight: Evaluating Research Idea Generation via Future Impact, Bo Jiang, 2026.
3. LLMs Can't Jump, Tom Zahavy (Google DeepMind), 2026.

Can AI automate engineering optimization tasks?

Can AI automate AI research itself?

Goal: enable agents to autonomously select ideas, run experiments, and produce validated research discoveries.

1. PaperBench: Evaluating AI's Ability to Replicate AI Research, Giulio Starace et al. (OpenAI), 2025.
2. Learning to Discover at Test Time, Mert Yuksekgonul et al. (Stanford University and NVIDIA), 2026.
3. FrontierCS: Evolving Challenges for Evolving Intelligence, Qiuyang Mang et al., 2025.
4. CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery, Ao Qu, Han Zheng, Zijian Zhou et al. (MIT and NUS), 2026.
5. EdgeBench: Unveiling Scaling Laws of Learning from Real-World Environments, Deyao Zhu et al. (ByteDance Seed), 2026.

How to effectively turn experience into persistent knowledge?

Goal: convert noisy interaction trajectories into reusable knowledge that persists across tasks without accumulating errors or catastrophic forgetting.

1. ExpeL: LLM Agents Are Experiential Learners, Andrew Zhao et al. (University of Maryland), 2024.
2. Agentic Context Engineering: Evolving Contexts for Self-Improving Language Models, Qizheng Zhang et al. (Stanford University), 2025.
3. Self-Distillation Enables Continual Learning, Idan Shenfeld et al. (MIT and Thinking Machines Lab), 2026.
4. Reinforcement Learning via Self-Distillation, Jonas Hübötter et al. (ETH Zurich and Stanford University), 2026.
5. Continual Learning Bench: Evaluating Frontier AI Systems in Real-World Stateful Environments, Asawa et al., 2026.

When should we update model weights, and when should we update the harness?

Goal: determine whether new lessons should be internalized into model weights or externalized via prompts, memory, skills, tools & orchestration.

1. GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning, Lakshya Agrawal et al. (UC Berkeley and Stanford University), 2025.
2. Meta-Harness: End-to-End Optimization of Model Harnesses, Yoonho Lee et al., 2026.
3. WHALE: A Simple Recipe for Joint Harness-Weight Optimization, Kim et al. (KRAFTON, KAIST & Stanford), 2026.
4. Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory, Chhikara et al. (Mem0), 2025.

How can we oversee increasingly autonomous systems?

Goal: maintain meaningful human control when agents operate over long horizons, exceed their supervisors in capability, and may learn to exploit or circumvent their evaluators.

1. AI Safety via Debate, Geoffrey Irving, Paul Christiano, and Dario Amodei (OpenAI), 2018.
2. Weak-to-Strong Generalization: Eliciting Strong Capabilities with Weak Supervision, Collin Burns et al. (OpenAI), 2023.
3. Agentic Misalignment: How LLMs Could Be Insider Threats, Anthropic, 2025.
4. CoT Red-Handed: Stress Testing Chain-of-Thought Monitoring, Benjamin Arnav et al. (Google DeepMind), 2025.

Self-Evolving AI for Science and Engineering

Goal: continually learn from simulations, experiments, and real-world feedback to accelerate scientific discovery and engineering optimization.

1. GeneBench-Pro: Evaluating Multistage Statistical Reasoning in Genomics, Quantitative Biology, and Translational Biomedicine, Jeremy Li and Andrew Ho (OpenAI), 2026.
2. Coscientist: Autonomous Chemical Research with Large Language Models, Boiko et al., 2023.
3. AnalogAgent: Self-Improving Analog Circuit Design Automation with LLM Agents, Zhixuan Bao et al., 2026.
4. Autonomous Agents Coordinating Distributed Discovery Through Emergent Artifact Exchange, Wang et al., 2026.

Self-Evolving AI for Personalization

Goal: continually learn from individual interactions and feedback to adapt to each user's evolving needs, preferences, and goals.

1. Learning Personalized Agents from Human Feedback, Kaiqu Liang et al., 2026.
2. OpenClaw-RL: Train Any Agent Simply by Talking, Yinjie Wang et al., 2026.
3. MIRIX: Multi-Agent Memory System for LLM-Based Agents, Yu Wang and Xi Chen, 2025.
4. StuLife: Building Self-Evolving Agents via Experience-Driven Lifelong Learning, ECNU-ICALK, 2026.

Self-Evolving AI for Multimodal & Embodied World

Goal: continually learn from multimodal perception and physical interaction to adapt their knowledge, skills, and behavior in dynamic real-world environments.

1. Self-Evolving Embodied AI, Tongtong Feng, Xin Wang, and Wenwu Zhu, 2026.
2. Voyager: An Open-Ended Embodied Agent with Large Language Models, Guanzhi Wang et al. (NVIDIA and Caltech), 2023.
3. Seeing, Listening, Remembering, and Reasoning: A Multimodal Agent with Long-Term Memory, Long et al., 2025.

Lecture Summary

- Course introduction and syllabus
- Background to LLMs, reasoning, and agents
- Introduction to self-evolving AI
- Research ideas and tasks for self-evolving AI

Assignments for This Coming Week

We will notify you about your enrollment status this week.

- For-credit enrollment will be capped at 20-30
- Listeners are welcome to join lectures, discussions and course projects if they are dedicated, but there will be limited space and resources

Project preference form will be out this week:

- Tentative list of teammates
- Tentative directions and tasks